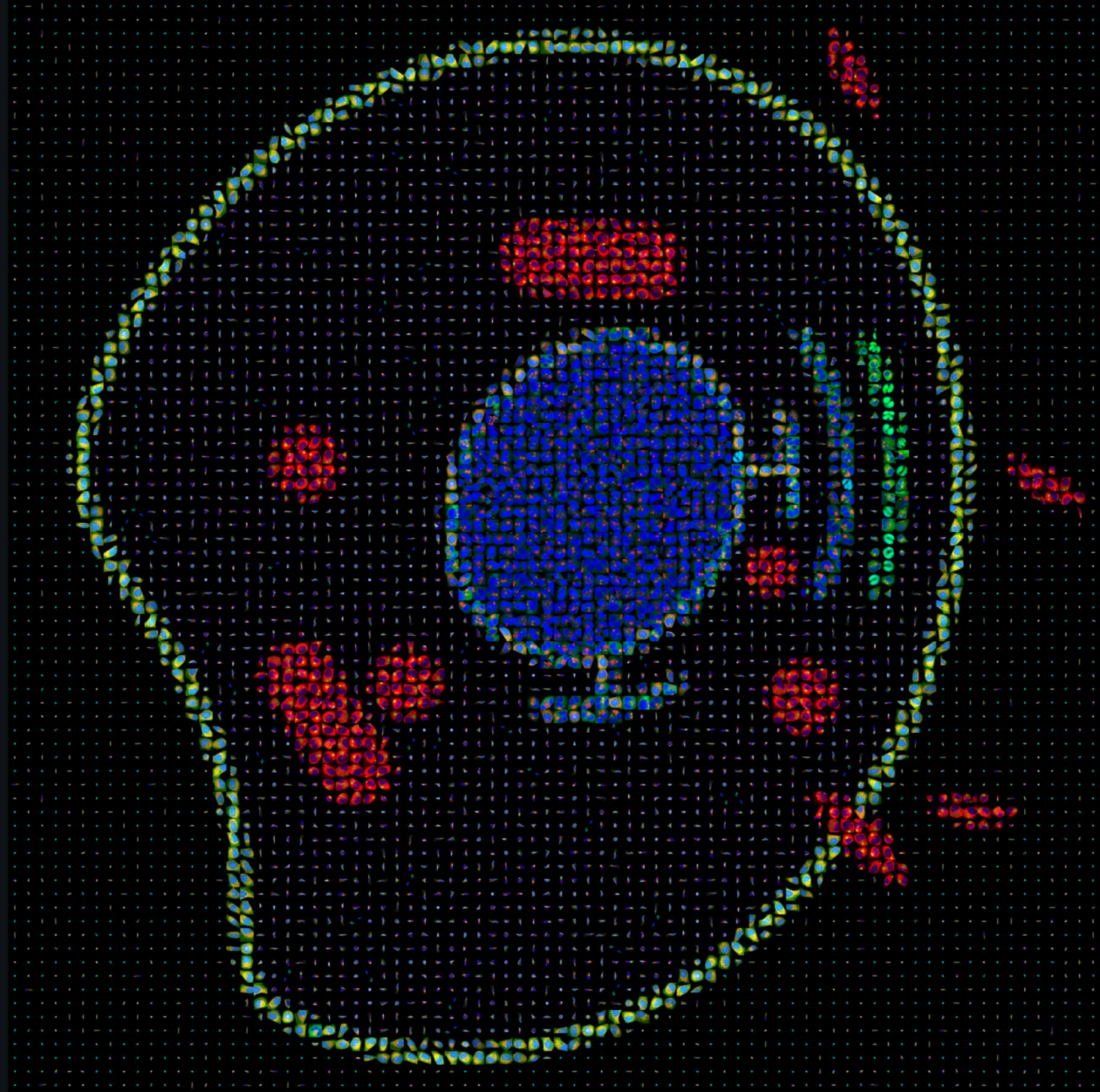
spaCR

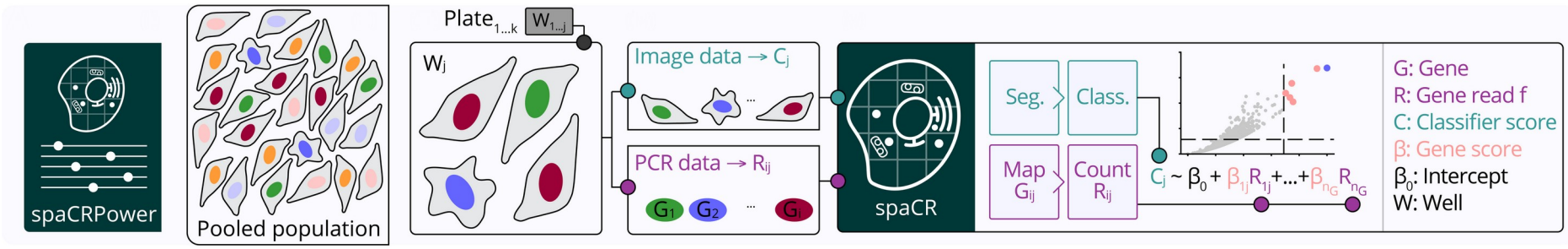
Spatial phenotype analysis of CRISPR screens



spaCR 1.5.1.4 · github.com/EinarOlafsson/spacr

INTRODUCTION

What is spaCR?



Pooled screen: every well holds a mix of knockouts. Image it, sequence it, and regress. (Preprint, Fig. 1A)

Image every well

Segment, measure and classify every cell. The phenotype becomes a number per well.

Sequence every well

Read the gRNA barcodes to learn which knockouts are in each well, and in what proportion.

Link the two

Regress the phenotype on guide composition. Out comes an effect per gene: a ranked hit list.

No sequencing? Segmentation, measurement, annotation, classification and the assays work on any high-content imaging experiment.

Where spaCR fits among existing tools

tool lacks it
tool has it: spaCR has it too
tool has it: spaCR does not

	CellProfiler	Fiji / ImageJ	napari	Cellpose	ilastik	QuPath	DL4Mic	MAGeCK	spaCR	Category
Best for	*cell measurement*	*image analysis*	*image viewing*	*segmentation*	*pixel / object ML*	*tissue slides*	*no-code DL*	*sequencing screens*	*image-based screens*	
Point-and-click interface	✓	✓	✓	✓	✓	✓	limited	✗	✓	Usability
Batch and headless runs (server, cluster)	✓	limited	✗	✓	✓	✓	limited	✓	✓	Automation
Reads most microscope file formats	✓	✓	plugin	limited	limited	✓	limited	✗	✓	Input
Deep-learning segmentation	plugin	plugin	plugin	✓	limited	plugin	✓	✗	✓	Segmentation
Train your own segmentation model	✗	plugin	plugin	✓	✓	limited	✓	✗	✓	Segmentation
Live assisted segmentation	✗	plugin	plugin	limited	✓	✓	✗	✗	✓	Segmentation
Correct masks by hand	limited	✓	✓	✓	✓	✓	✗	✗	✓	Segmentation
Per-cell measurements	✓	✓	plugin	✗	✓	✓	✗	✗	✓	Measurement
3-D (z-stack) images	✓	✓	✓	✓	✓	limited	limited	✗	beta	Image types
Time-lapse tracking	✓	plugin	plugin	✗	✓	✗	✗	✗	✓	Image types
Whole plates in one run	✓	limited	✗	limited	limited	limited	limited	✗	✓	Screening
Plate maps, heat maps and QC	limited	✗	✗	✗	✗	✗	✗	✗	✓	Screening
Classify cells from images (deep learning)	✗	✗	✗	✗	✗	✗	✗	✗	✓	Classification
Classify cells from measurements	limited	✗	plugin	✗	✓	✓	✗	✗	✓	Classification
Explore cells: UMAP with images, gating	limited	✗	plugin	✗	✗	✗	✗	✗	✓	Analysis
Plots and statistics without code	limited	limited	plugin	✗	✗	limited	✗	limited	✓	Analysis
GPU acceleration	✗	plugin	plugin	limited	limited	limited	✓	✗	✓	Performance
Pooled CRISPR screens: images to genes	✗	✗	✗	✗	✗	✗	✗	limited	✓	Screening
Pixel classifier from brush strokes	✗	plugin	plugin	✗	✓	✓	✗	✗	✗	Segmentation
Whole-slide / tissue images	✗	limited	✗	✗	✗	✓	✗	✗	✗	Image types
Large plugin ecosystem	limited	✓	✓	✗	✗	limited	✗	✗	✗	Extensibility

Core features; 'plugin' = needs an add-on, 'limited' = partly. Simplified: check each tool for details.

INTRODUCTION

ALPHA

Alpha features: switch them on in Preferences

Category	Alpha feature
Segmentation	StarDist segmentation backend
	InstanSeg segmentation backend
	Omnipose segmentation backend
	DINOCell segmentation backend
	SAMCell segmentation backend
	Spotiflow spot detection
	SpotNet spot detection
	Click- and box-prompted segmentation (micro-SAM)
	Segmentation uncertainty maps
	Parameter-robustness report
	Run CellProfiler .cppipe pipelines
	Masks and ROIs in and out: QuPath, ImageJ, COCO
	Virtual staining from brightfield
	Mask batches across several GPUs
	Use Make Masks settings in Mask Generation
	MitoNet: mitochondria in TEM images
Image preprocessing	Spectral unmixing (bleed-through correction)
	Photobleaching correction
	Vendor flat-field profiles (Harmony, ZEN)
	Intensity calibration across sessions
	Deep-learning image QC: focus, debris, empty fields
	Self-supervised denoising (Noise2Void), low-light benchmark
	Illumination, enhancement and PSF settings
Measurements and assays	Cell-cycle phase classification
	Live/dead viability and cytotoxicity index
	Confluency for any channel
	Scratch / wound-closure assay
	Colony / CFU counting
	Image-based profiling: normalise, select, mAP
	GPU measurement (cuCIM, CuPy)
	Estimate plaque scale and growth time

Category	Alpha feature
Time-lapse	SAM2 propagation: segment once, follow each object
	Timeflows: a temporal Cellpose
	Timepose v1: temporal Cellpose-SAM tracking
	Lineage trees from tracks
	Event detection: mitosis, egress, invasion, death
	Time-to-event: Kaplan-Meier and Cox
Cells and machine learning	Foundation-model embeddings (SubCell, OpenPhenom, …)
	Well labels to responding cells (multiple-instance)
	Anomaly detection against negative controls
	Find cells like this one (similarity search)
	Blind scoring
	Counterfactuals: morph a control cell toward a hit
	Self-supervised DINO pretraining
	Judge every label from cross-channel ground truth
Screens and statistics	Arrayed-screen hits: SSMD, robust z, B-score
	Chemistry-aware hits: SMILES, clusters, SAR
	Sample-size planner for arrayed assays
	Spatial transcriptomics: Visium, Xenium; spatial masks, barcodes
	LIMS / plate-barcode linkage
Automation and data	Watch a folder, analyse images as they arrive
	Smart microscopy: send hits back for re-imaging
	Cloud storage (S3 and others)
	Export a run as a Nextflow / Snakemake workflow
	Run-finished notifications (email, Slack, ntfy)
	Export for R (SingleCellExperiment), tidy Parquet
	Plugin and recipe catalogue
	Postgres or DuckDB / Parquet measurement store
	Test data on seven more modules
	Export masks and objects for vvvv
Reproducibility	Figure-integrity check
	Pre-registered analysis lock
	Metadata and archive packages (REMBI, IDR)
	One-click Zenodo archive with a DOI

Alpha species	Plasmodium	Candida	Trypanosoma	Leishmania	Giardia	Virus	Mammalian

64 alpha features and 7 alpha species, hidden until Preferences → Modules → Show alpha features and Show alpha species are on. Early versions: expect changes.

INTRODUCTION

Six steps from plate to hit list

1

Mask

Find every cell, nucleus, parasite

2

Measure

Shape, intensity, texture per object

3

Annotate

Label example cells

4

Classify

Score every cell's phenotype

5

Map Barcodes

Which gRNAs are in which well

6

Regression

Which genes change the phenotype

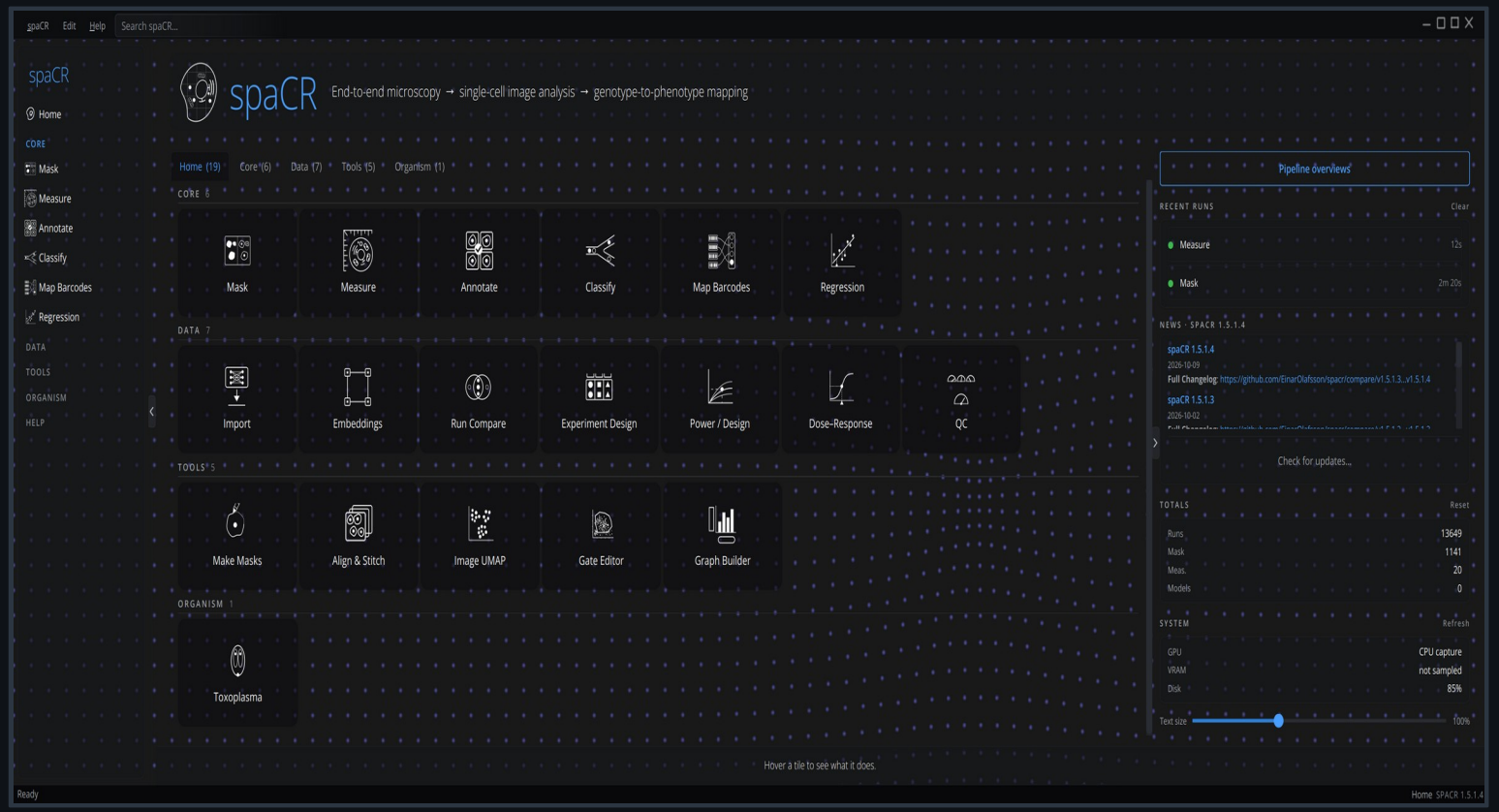
images

sequencing + statistics

One project, one database. Images, masks, crops, measurements, annotations, predictions, barcodes and well IDs all live in one SQLite file (measurements.db). Every module reads from it and writes back to it, and every run saves its settings so it can be replayed.

INTRODUCTION

19 modules in one desktop app



The home screen: every module is a tile, grouped by what it is for

6 **Core**
Mask · Measure · Annotate · Classify · Map Barcodes · Regression

7 **Data**
Import · Embeddings · Run Compare · Experiment Design · Power · Dose–Response · QC

5 **Tools**
Make Masks · Align & Stitch · Image UMAP · Gate Editor · Graph Builder

1 **Organism**
Toxoplasma: one module per organism, its assays inside. More organisms are alpha species

Plus ~40 sub-tools inside the modules. Ctrl+K opens a command palette that reaches every one of them.

INTRODUCTION

Runs on a laptop, a workstation or a cluster

Desktop app

One borderless window that opens with a logo animation (replay it from Help). Live previews, a console per module, settings search, recipes, and a Run button.

Headless

The same modules and settings files run without the window, on a server or a cluster. Export settings… on every form saves it as a settings CSV for spacr-run.

Reproducible

Every run records its settings. spacr-repro RUN_DIR replays it. Run History and Run Compare show what changed.

4

kinds of graphics card

NVIDIA, AMD, Apple and Intel: or just the CPU

10

interface languages

switch in Preferences without restarting

84

interactive tutorials

narrated videos from install to hit investigation

3.9–3.14

Python versions

3.12 recommended; Linux, macOS, Windows

Accessibility Preferences → Appearance → Accessibility: High contrast, Text size, Reduce motion, stronger focus outlines and a keyboard-shortcut editor. Screen readers announce every control, in the interface language.

INTRODUCTION

Install spaCR

Desktop installer

Easiest. Bundles its own Python: no conda needed.

```
GitHub → Releases → latest
Windows   .exe
macOS     .pkg   (Intel + Apple silicon)
Linux     chmod +x SpaCR-*.run && ./SpaCR-*.run
```

pip (conda env)

Most flexible; add optional extras.

```
conda create -n spacr python=3.12 -y
conda activate spacr
pip install "spacr[qt]"
spacr
```

conda-forge

One command into an existing env.

```
conda create -n spacr python=3.12 -y
conda activate spacr
conda install conda-forge::spacr
spacr
```

From source

For development (nightly branch).

```
git clone --branch nightly \
  https://github.com/EinarOlafsson/spacr
cd spacr && pip install -e .
spacr
```

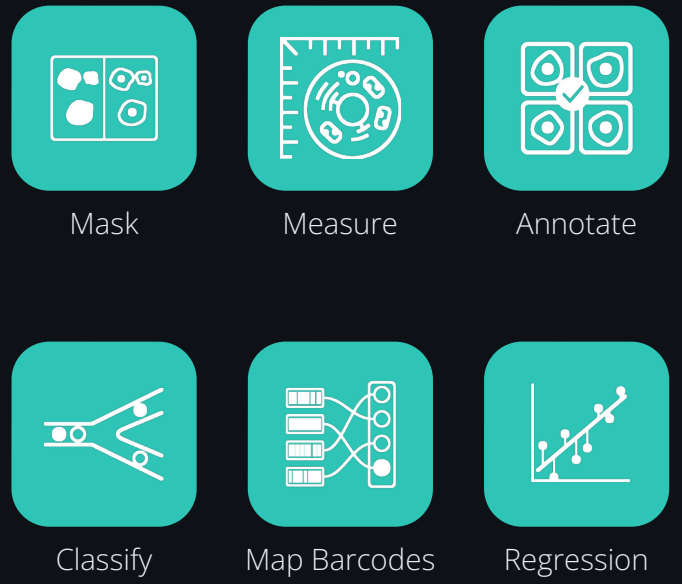
Update and remove Help → Check for updates installs the new version in place of the old one and keeps your data. Preferences → Performance → Uninstall… lists the installed versions and the categories of data to remove.

Something off? `spacr-doctor` checks the install and says how to fix it. Headless server: `pip install spacr` (no Qt).

01

Core pipeline

From raw plate images and FASTQ reads to a ranked list of genes.



CORE · 1 OF 6

Mask: segmentation

STABLE

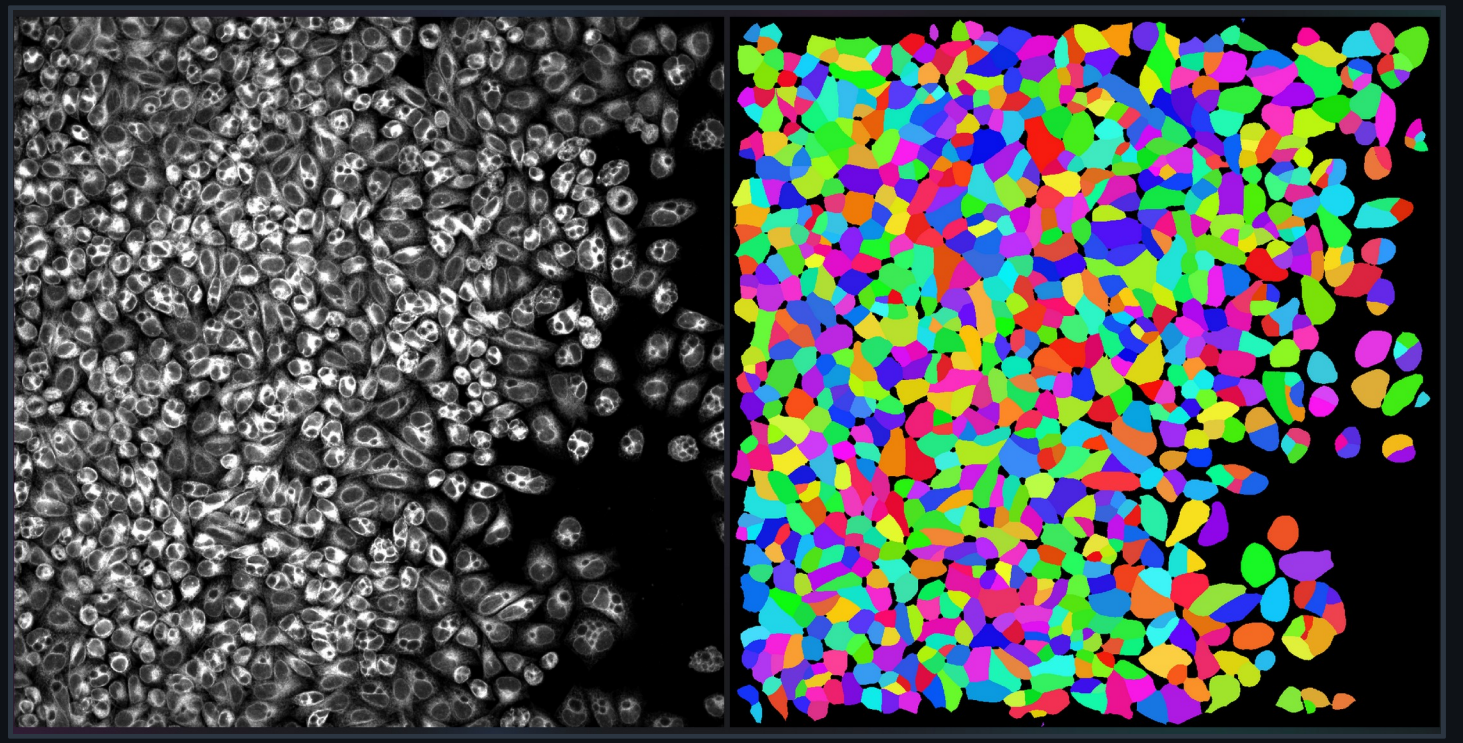
Finds and outlines every cell, nucleus, parasite and organelle in your images.

WHAT YOU CAN DO

- Reads files from most microscopes (TIFF, Leica, Zeiss, Nikon)
- Cellpose-SAM by default; other segmentation models optional
- Cell, nucleus, pathogen and up to 26 organelle classes
- Plug in fine-tuned Model Zoo models (e.g. Toxoplasma vacuoles)
- 2-D, 3-D z-stacks and time-lapse; live preview before a full run

OUTPUT

- One mask per object type, aligned to the images
- Overlay images and a quality score per field
- The settings used, saved with the run

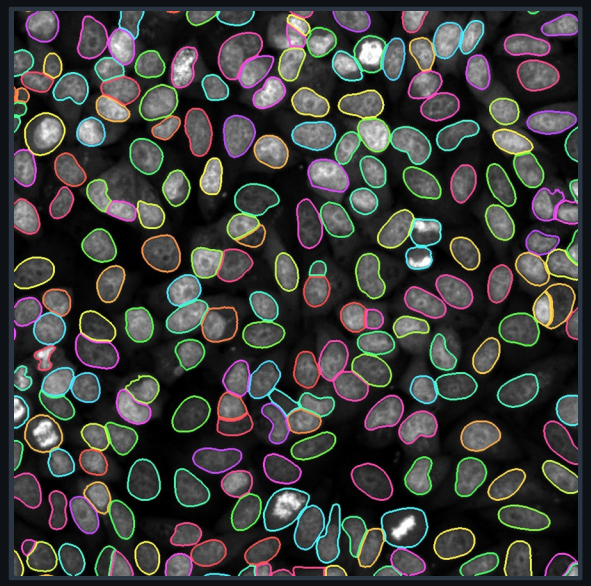


Live preview: host cells (left) and the label mask Cellpose-SAM returns (right)

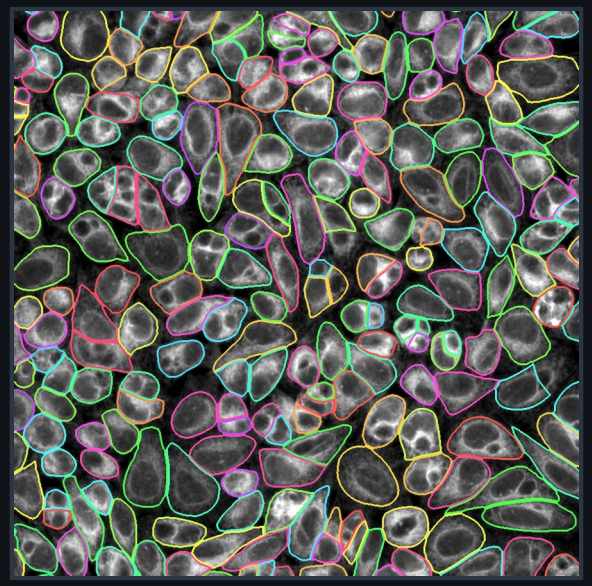
CORE · 1 OF 6

Mask: every object class

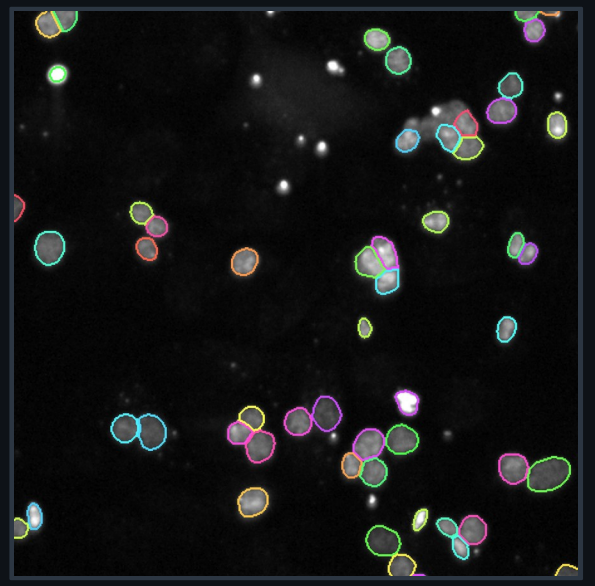
STABLE



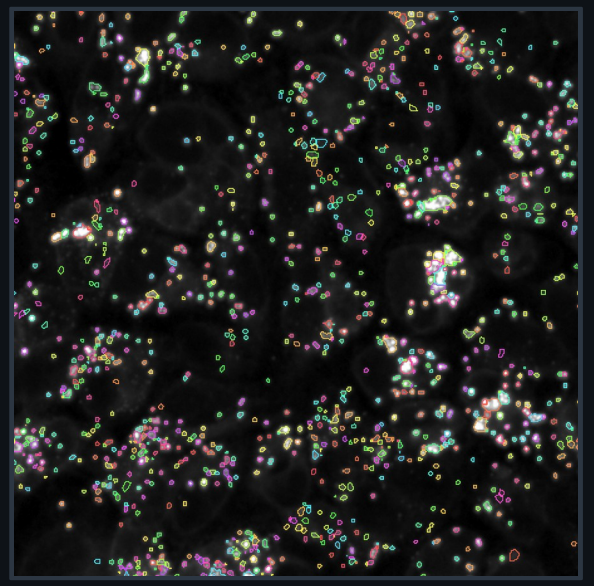
Nuclei: Cellpose-SAM (Hoechst)



Cells: Cellpose-SAM (cell stain)



Pathogen: Toxoplasma, zoomed



Organelles: lipid droplets, zoomed

SUB-TOOLS INSIDE THIS MODULE

Timelapse
Follow each cell and parasite from frame to frame in movies

Cellpose flows
See why two cells were merged, or one was split

CORE · 2 OF 6

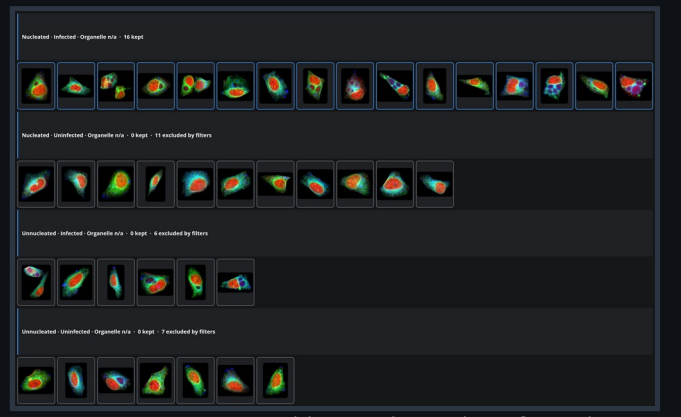
Measure: features and crops

STABLE

Turns every segmented object into a row of numbers: shape, brightness, texture and position.

WHAT YOU CAN DO

- Shape, size, brightness and texture of every object
- Position: neighbours, distances, touching the edge
- Channel overlap; measure inside a chosen region
- Saves object crops for Annotate and Classify
- 3-D objects measured in µm
- Checks free RAM first and sizes its workers to fit



Crop preview, grouped by nucleated / infected state

The cell table in the Database Browser

OUTPUT

- One table per object type (cells, nuclei, parasites, …)
- Per-object PNG crops

SUB-TOOLS INSIDE THIS MODULE

Illumination
Flat-field correction estimated from the plate itself

AnnData export
Export for single-cell analysis tools (scanpy)

Motility assay
Track velocity and straightness, split by infection state

Database Browser
Browse, filter and export any project table

CORE · 3 OF 6

Annotate: label cells by hand

STABLE

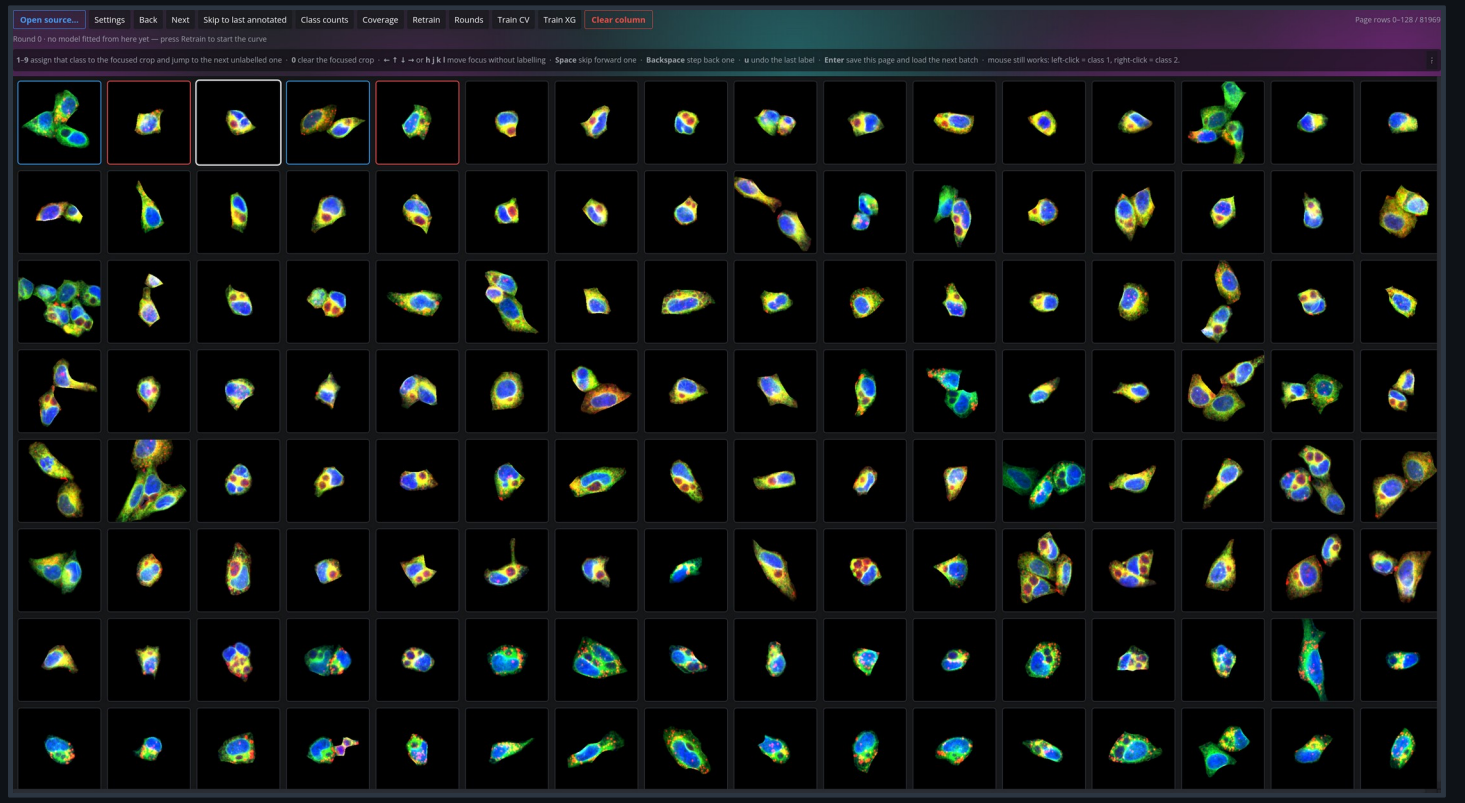
Label single-cell crops so a classifier can learn what each phenotype looks like.

WHAT YOU CAN DO

- Pages through a grid of crops from the project
- Click, or press 1–9 with auto-advance; 0 clears, u undoes
- Several annotation columns side by side (e.g. two annotators)
- Active learning: shows the cells the model is least sure about first
- Change channels and colours for display only; files stay untouched

OUTPUT

- Labels in an annotation column of the project
- Class counts and coverage per well



Annotation grid. Selected crops are outlined; the queue can be ranked by model uncertainty

SUB-TOOLS INSIDE THIS MODULE

Annotator Agreement
How often two people label a cell the same way: and a list of the cells they disagreed on

CORE · 4 OF 6

Classify: score every cell

ALPHA

Teach the computer what a phenotype looks like, then let it score every cell in the screen.

WHAT YOU CAN DO

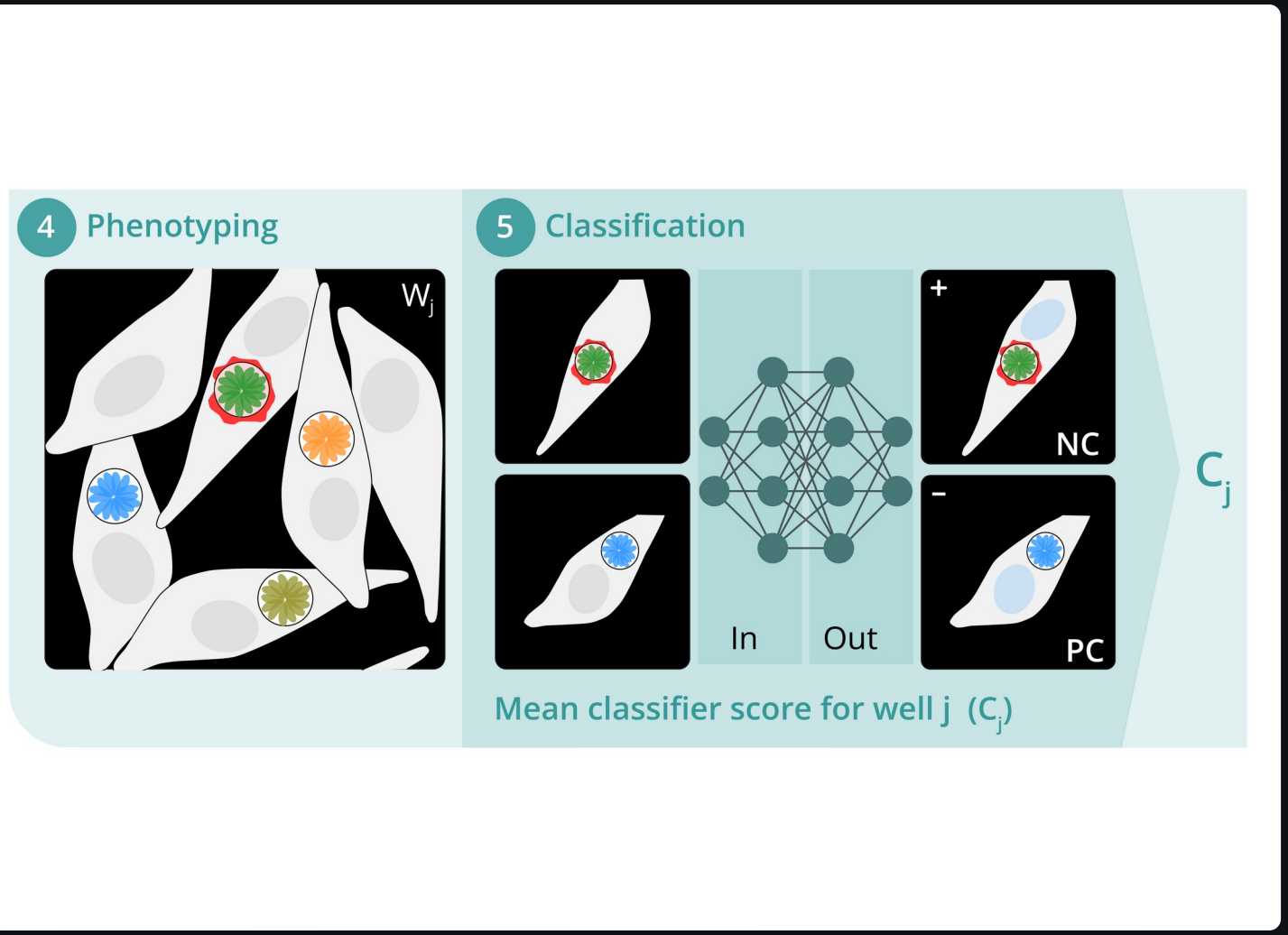
Image model: learns straight from the cell pictures (ResNet, EfficientNet, ViT, MaxViT, …)

Measurement model: learns from the measured features (XGBoost, random forest, …)

- Examples come from control wells or from your Annotate labels
- Tested on wells it never trained on, so the accuracy is honest
- Scores every cell and writes the result back to the project

OUTPUT

- A class and a probability for every cell
- The mean score per well → input to Regression

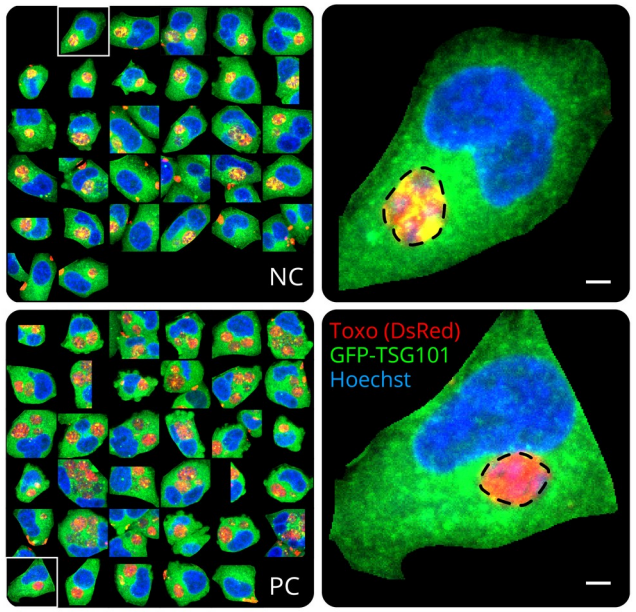


Each cell is scored by the classifier; a well's score is the mean of its cells (preprint Fig. 2)

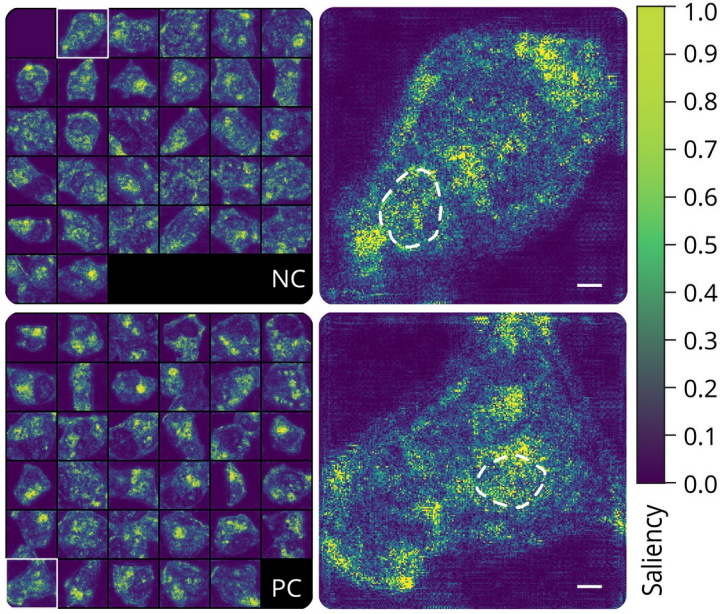
CORE · 4 OF 6

Classify: image model

ALPHA



Input: single-cell crops from negative (NC) and positive (PC) control wells



Saliency: where MaxViT looked: at the vacuole in PC cells

SUB-TOOLS INSIDE THIS MODULE

Activation Maps
Show which pixels drove each decision

Classifier Evaluation
Accuracy and mistakes per class and per plate, on unseen wells

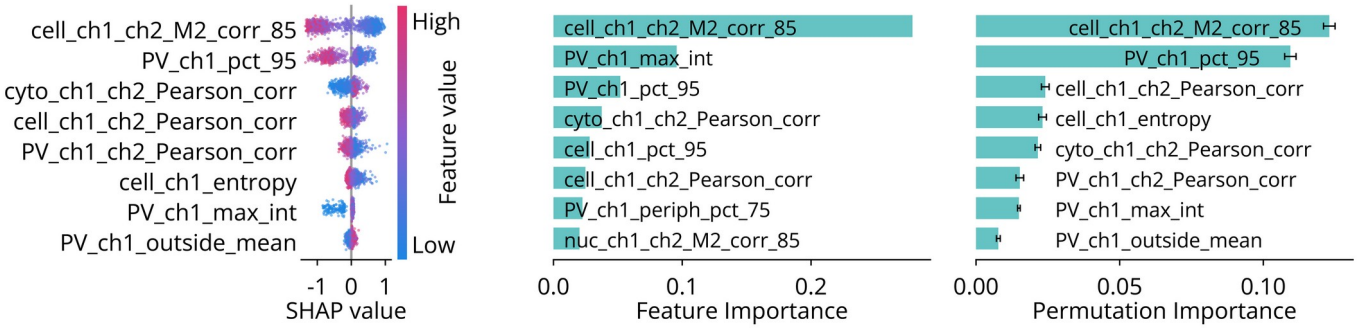
Training Runs
Compare training curves between runs

Explain CV Model
Explain an image model in terms of measurements

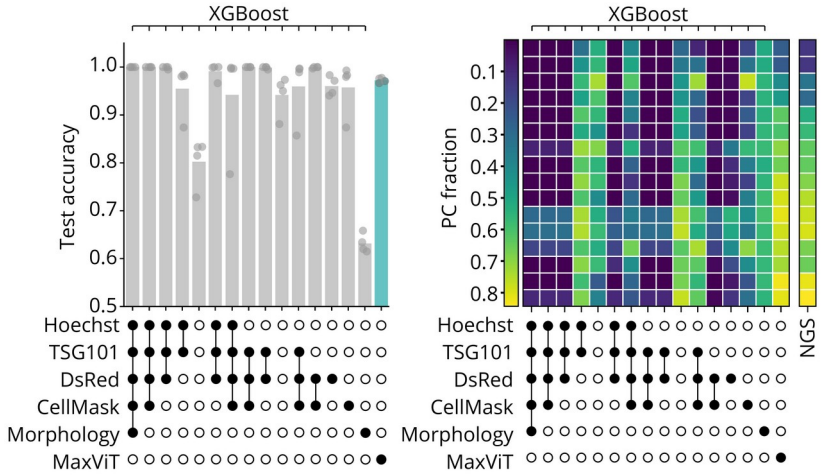
Feature Explorer
Which measurements separate the classes best

Classify: measurement model

ALPHA



XGBoost on the measurements: which measurements decide the class, three ways of ranking them



Accuracy with different channel combinations (grey, XGBoost) vs the image model (teal, MaxViT)

CORE · 5 OF 6

Map Barcodes: which guides, which well

STABLE

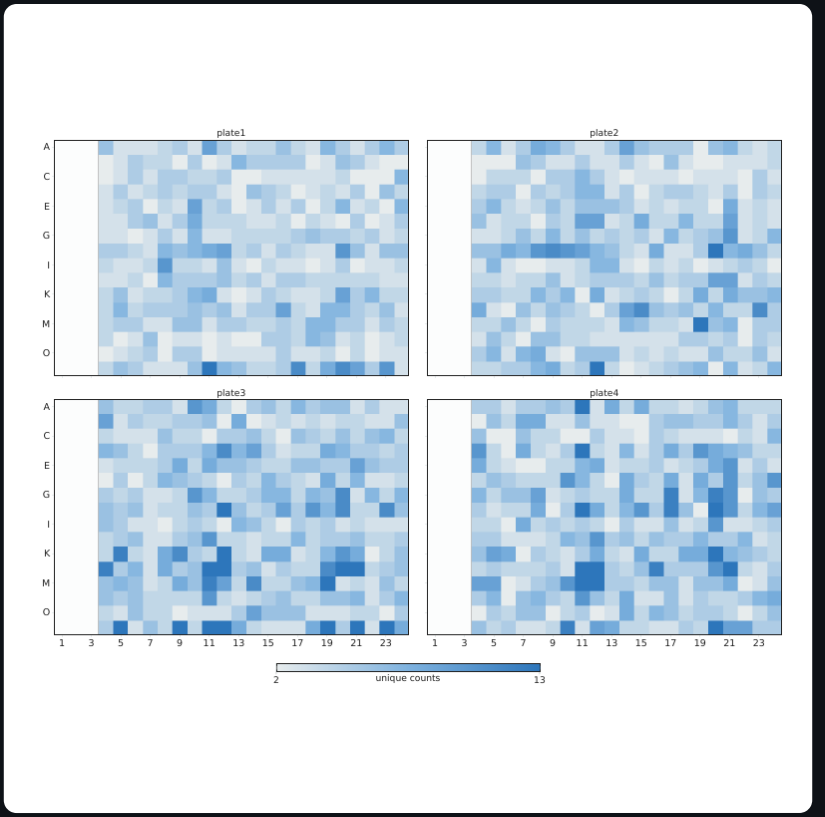
Reads the sequencing data and works out which gRNAs are in each well, and how much of each.

WHAT YOU CAN DO

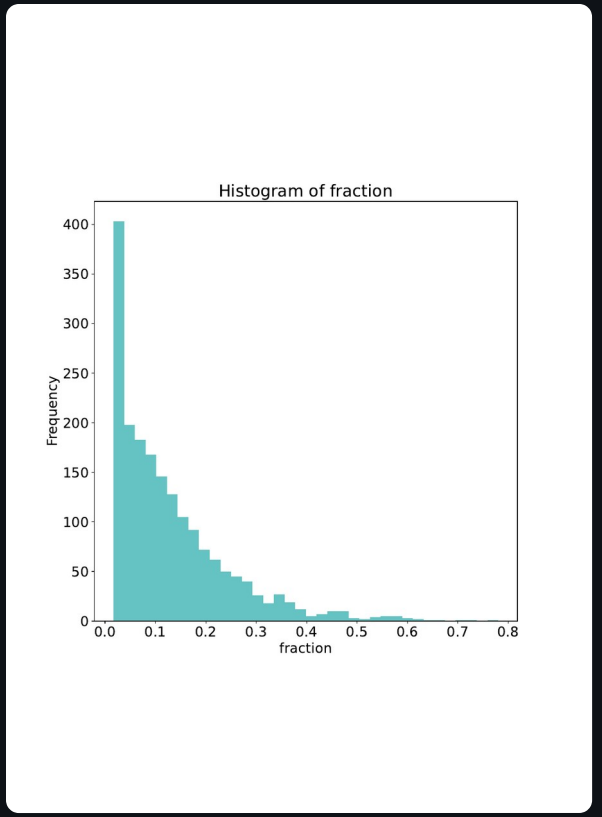
- Reads the row, column and gRNA barcodes from the sequencing files
- You give it the barcode lists and where they sit in the read
- Paired-end or single reads; set how many mismatches to allow
- Quick test on a small part of the file first

OUTPUT

- Guide counts per well
- QC table and run manifest
- gRNA fraction per well → input to Regression



Unique gRNAs per well, four plates



gRNA fraction per well

SUB-TOOLS INSIDE THIS MODULE

Barcode QC
Reads per well, reads that did not match, library coverage: and the cut-off that gives the expected number of gRNAs per well

CORE · 6 OF 6

Regression: from wells to genes

STABLE

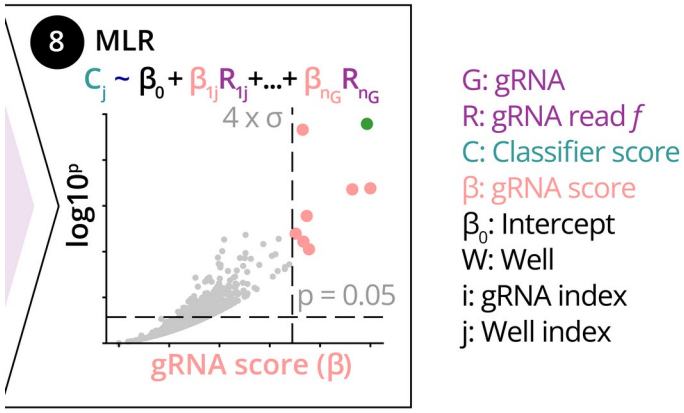
Works out which genes change the phenotype, and ranks them.

WHAT YOU CAN DO

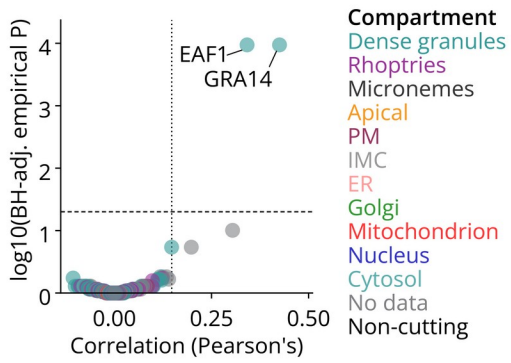
- Pairs each well's score with the fraction of each gRNA in that well
- Default: a mixed model where the gRNAs of a gene are grouped
- Other models for counts, fractions or skewed data
- Permutation test: shuffles gRNAs to check a hit is not chance

OUTPUT

- A ranked table of genes with effect size and p-value
- Volcano, top effects and quality-check plots



The model: a well's score = its gRNA fractions × each gene's effect (preprint Fig. 2)



The result: EAF1 and GRA14 stand out (preprint Fig. S7)

CORE · 6 OF 6

Regression: choose the model

STABLE

21 model families, each explained in plain words, with what it assumes about your data.

WHAT YOU CAN DO

Mixed: the gRNAs of a gene grouped, genes allowed to vary

OLS / WLS: classic least squares, optionally weighted by cells per well

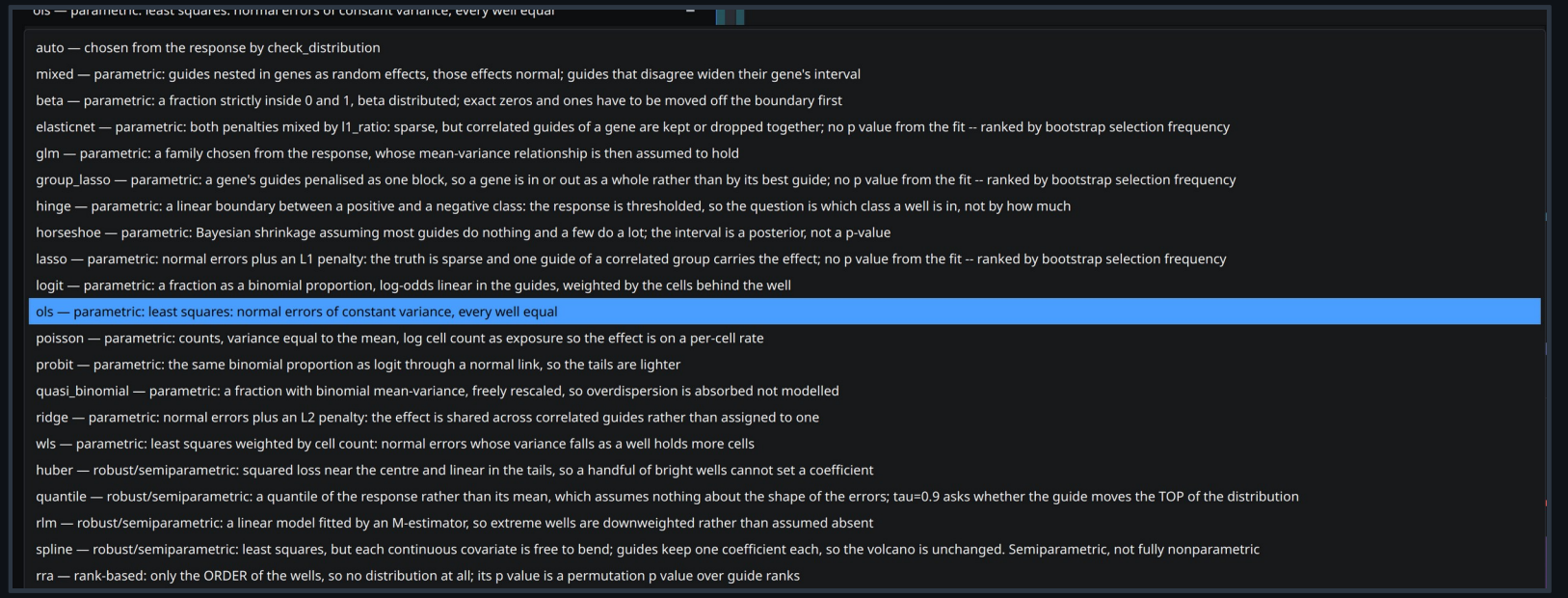
GLM, Poisson, beta, logit: for counts, fractions and yes/no readouts

Lasso, ridge, elastic net, group lasso, horseshoe: when there are many more genes than wells

Robust, quantile, RRA: when a few wells are extreme

OUTPUT

- Plate, row and column covariates; permutation test on gRNAs
- The formula and what it assumes, shown for the chosen model

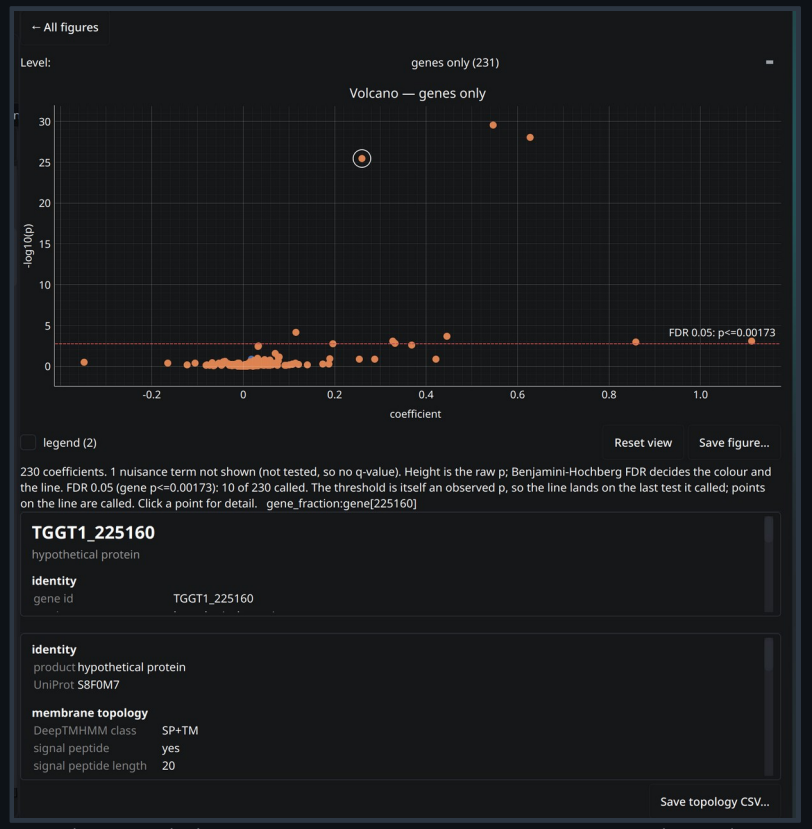


Regression type: every model with the assumption it makes (OLS selected)

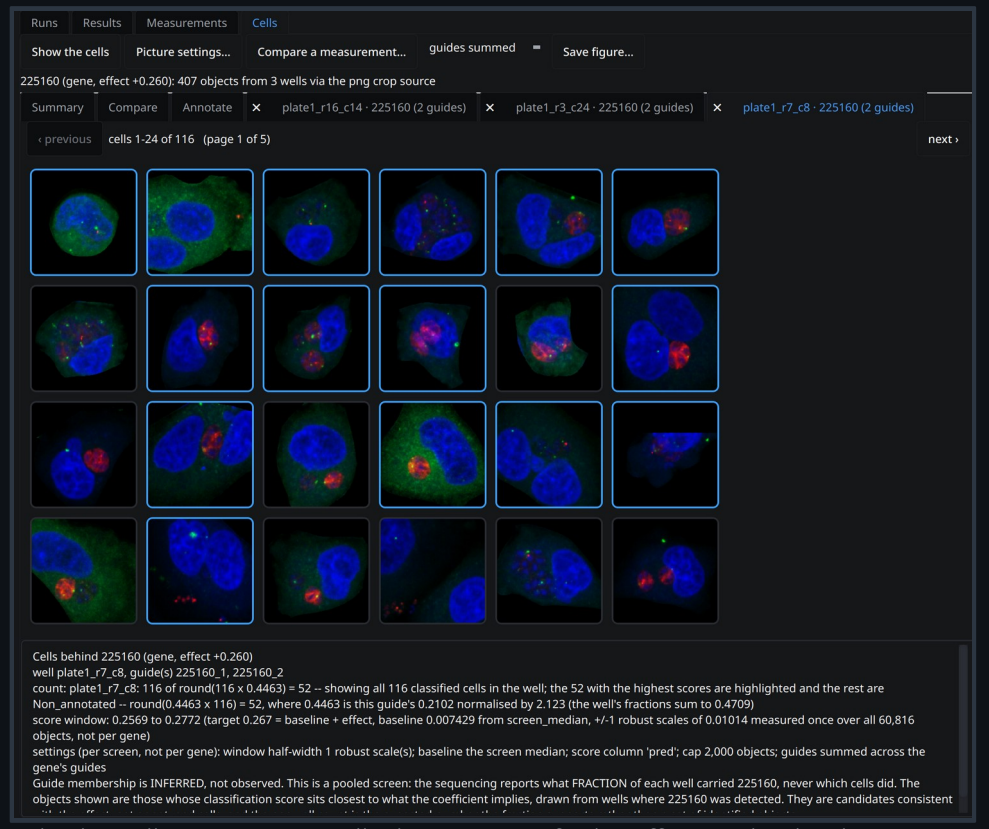
CORE · 6 OF 6

Regression: pick a gene, see its cells

STABLE



Interactive volcano: click a point. EAF1 (TGGT1_225160) is selected, its gene card below



Cells tab: the cells in an EAF1 well whose scores fit the effect, with why they were picked

SUB-TOOLS INSIDE THIS MODULE

Volcano Explorer
Publication version of the volcano; vector export

Investigate Hit
Test the hit at the cell level against control wells

Hit List
Ranked genes with intervals and gRNA agreement

Prediction Profiler
How the prediction changes as one gene's fraction changes

CORE · 6 OF 6

Regression: assumption checks

STABLE

Every run checks its own assumptions and says, in red, which ones fail and what to do.

WHAT YOU CAN DO

Equal variance: Breusch-Pagan test on the residuals

Normal residuals: skew, kurtosis, D'Agostino test

Independence: Durbin-Watson; wells on one plate share handling

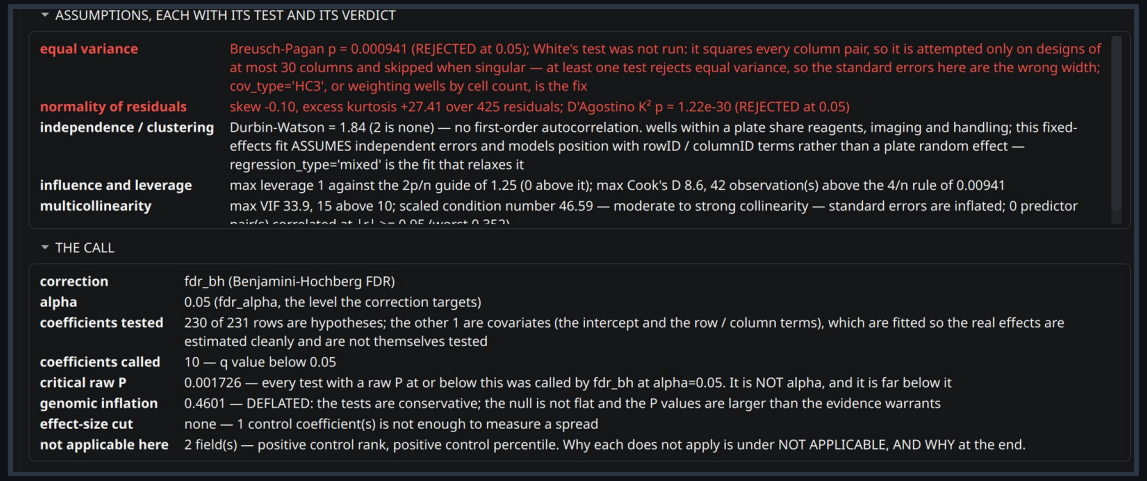
Influence: leverage and Cook's distance per well

Collinearity: VIF and condition number

Calibration: p-value histogram and genomic inflation

OUTPUT

- A run summary: the answer, what was fitted, each assumption with its verdict
- Residual, Q-Q, scale-location and influence plots



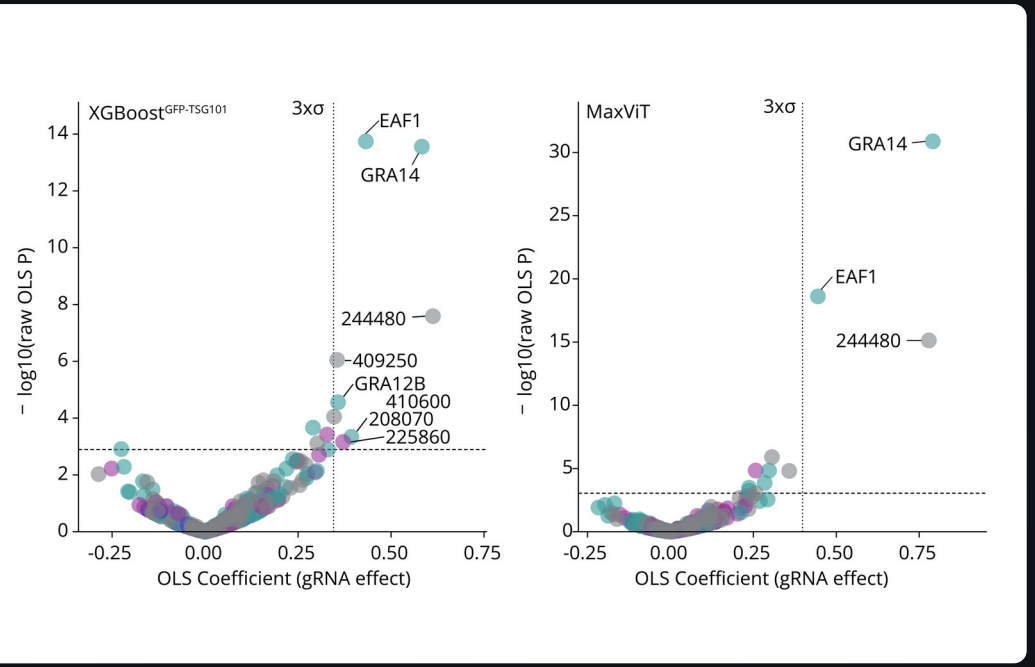
Summary tab: each assumption with its test and verdict; equal variance and normality rejected (red)

The diagnostic plots behind the verdicts

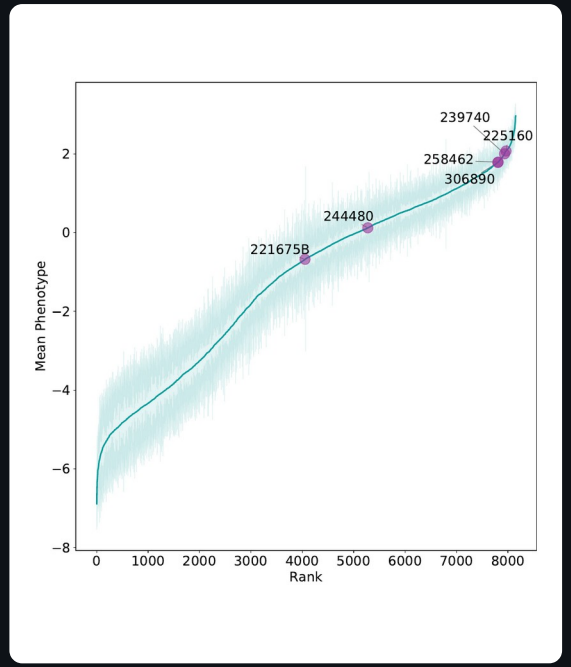
CORE · 6 OF 6

Regression: the hit list

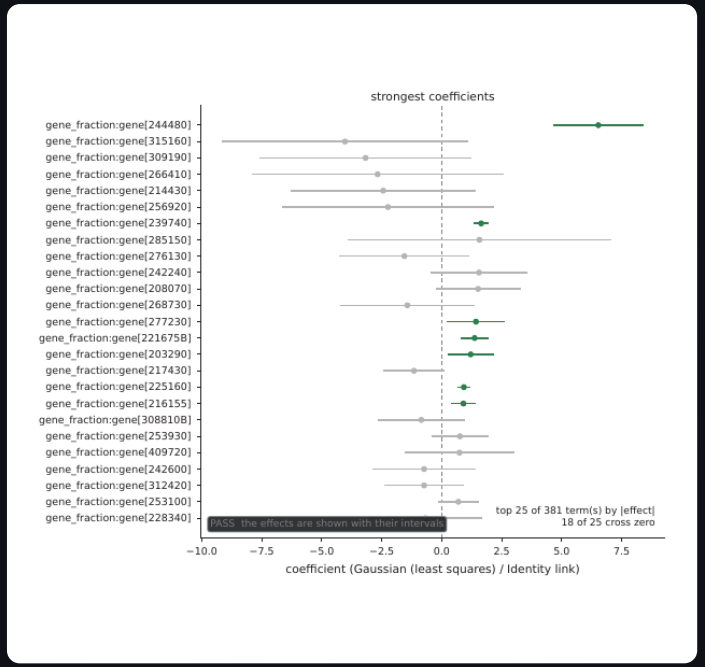
STABLE



Volcanos from the screen: XGBoost (left) and MaxViT (right) scores both find EAF1 and GRA14 (preprint Fig. 5)



Genes ranked by phenotype



Strongest effects with 95% intervals

SUB-TOOLS INSIDE THIS MODULE

Volcano Explorer
Click a point to see the gene; export for publication

Hit List
Ranked genes with confidence intervals and gRNA agreement; export as a table

Investigate Hit
Jump from a gene to its wells and cells

Diagnostics
Checks that the model fits the data (plate, row and column effects)

Methods & Results
Drafts the methods text from the run itself

Prediction Profiler
How a prediction changes across one input

Parameter Sweep
Re-run across settings to see how stable the hits are

02

Data

Bring data in, plan experiments, and check that a run can be trusted.

Import

Embeddings

Run Compare

Experiment Design

Power / Design

Dose–Response

QC

DATA · 1 OF 7

Import: bring outside data in

STABLE

Turns images, masks and tables from other tools into a spaCR project.

WHAT YOU CAN DO

- Pair images, instance masks and a measurement CSV
- Match their column names to spaCR's, converting units
- Choose what happens when names collide
- Preview before anything is written; save and reuse the mapping

OUTPUT

- A spaCR project that records where the data came from

Their column	spaCR column	Transform	Unit in	Unit out	Note
AreaShape_Area_um2	foreign_areashape_area_um2	area	um^2	px^2	unit um^2 read from the column name; spaCR measures in px^2, so this needs a pixel size
MeanIntensity_ER	foreign_meanintensity_er	identity			copied unchanged; declare unit_in/unit_out if it needs scaling
Metadata_Well	foreign_metadata_well	identity			copied unchanged; declare unit_in/unit_out if it needs scaling
cell_area	foreign_cell_area	identity			name suggests a area, but no unit is declared — set unit_in (and transform=area) if these are not already in px^2

Column mapping: their name → spaCR column, with unit conversion

Source	Target	Plate	Well	Field	Channel	Z	T	From well	From field	From channel	Z handling	Status
fov01_C1.tif	plate1_A01_T0001F001L01A01Z01C01.tif	plate1	A01	1	1	1	1	A01	images/fov01	C1	keep	planned
fov01_C2.tif	plate1_A01_T0001F001L01A01Z01C02.tif	plate1	A01	1	2	1	1	A01	images/fov01	C2	keep	planned
fov01_C3.tif	plate1_A01_T0001F001L01A01Z01C03.tif	plate1	A01	1	3	1	1	A01	images/fov01	C3	keep	planned
fov01_C4.tif	plate1_A01_T0001F001L01A01Z01C04.tif	plate1	A01	1	4	1	1	A01	images/fov01	C4	keep	planned
fov02_C1.tif	plate1_A01_T0001F002L01A01Z01C01.tif	plate1	A01	2	1	1	1	A01	images/fov02	C1	keep	planned
fov02_C2.tif	plate1_A01_T0001F002L01A01Z01C02.tif	plate1	A01	2	2	1	1	A01	images/fov02	C2	keep	planned
fov02_C3.tif	plate1_A01_T0001F002L01A01Z01C03.tif	plate1	A01	2	3	1	1	A01	images/fov02	C3	keep	planned
fov02_C4.tif	plate1_A01_T0001F002L01A01Z01C04.tif	plate1	A01	2	4	1	1	A01	images/fov02	C4	keep	planned

Format Converter: plan for turning vendor files into spaCR's TIFF layout

SUB-TOOLS INSIDE THIS MODULE

Import Images
Reads any microscope's folder and works out the naming itself

Format Converter
Turns Nikon, Zeiss and Leica files into spaCR's layout

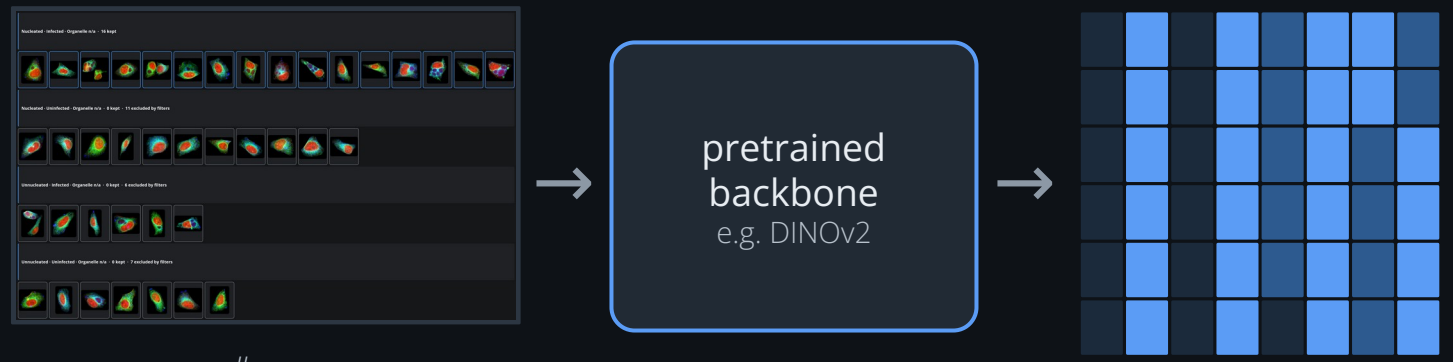
External Masks
Measure masks made elsewhere, without re-segmenting

DATA · 2 OF 7

Embeddings: one feature vector per cell

ALPHA

Gives every cell a label-free feature vector from a pretrained image model, next to the measured features.



cell crops

one row of numbers per cell

WHAT YOU CAN DO

- Crops from the project (filter by object class or plate) or from a folder
- Choose the network: DINOv2 (default; Cell-DINO when installed), ResNet-18, ResNet-50, ConvNeXt, ViT
- Each stain is described separately, or all together
- Runs on the graphics card when there is one

OUTPUT

- One row of numbers per cell, next to its measurements
- Which network was used is recorded

Use it for clustering and UMAPs without choosing features, finding cells that look like a hit, and checking whether a classifier is picking up something the measurements miss.

DATA · 3 OF 7

Run Compare: differences between runs

BETA

Puts two runs of the same project side by side, so you know why the results moved.

WHAT YOU CAN DO

- Pick a project and any two of its runs
- Settings diff, grouped by settings panel
- Object, well and field counts per plate
- Hits that appeared, vanished or changed rank
- Incompatible runs are blocked unless you choose 'compare anyway'

OUTPUT

- Side-by-side comparison tables

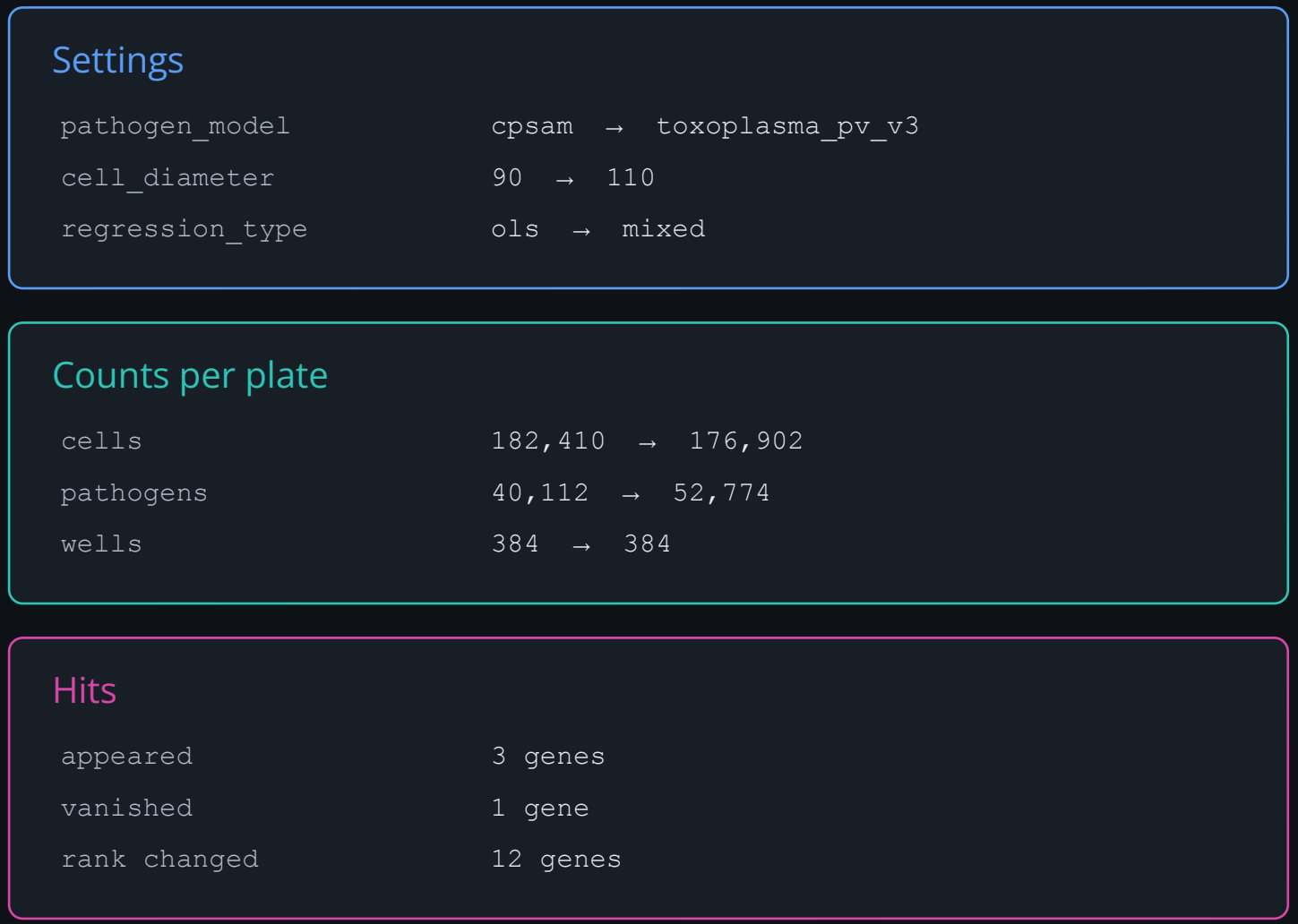


Illustration of the three comparison panels (values made up)

DATA · 4 OF 7

Experiment Design: plan the plate

ALPHA

Lay out conditions, controls and replicates before you image anything.

WHAT YOU CAN DO

- Assign conditions and control roles to wells
- Warnings for capacity, too few replicates, and row / column / edge confounding
- Randomise with a seed, so the layout can be rebuilt exactly

OUTPUT

- A plate map that later spaCR steps use directly
- The settings used, so the layout can be rebuilt

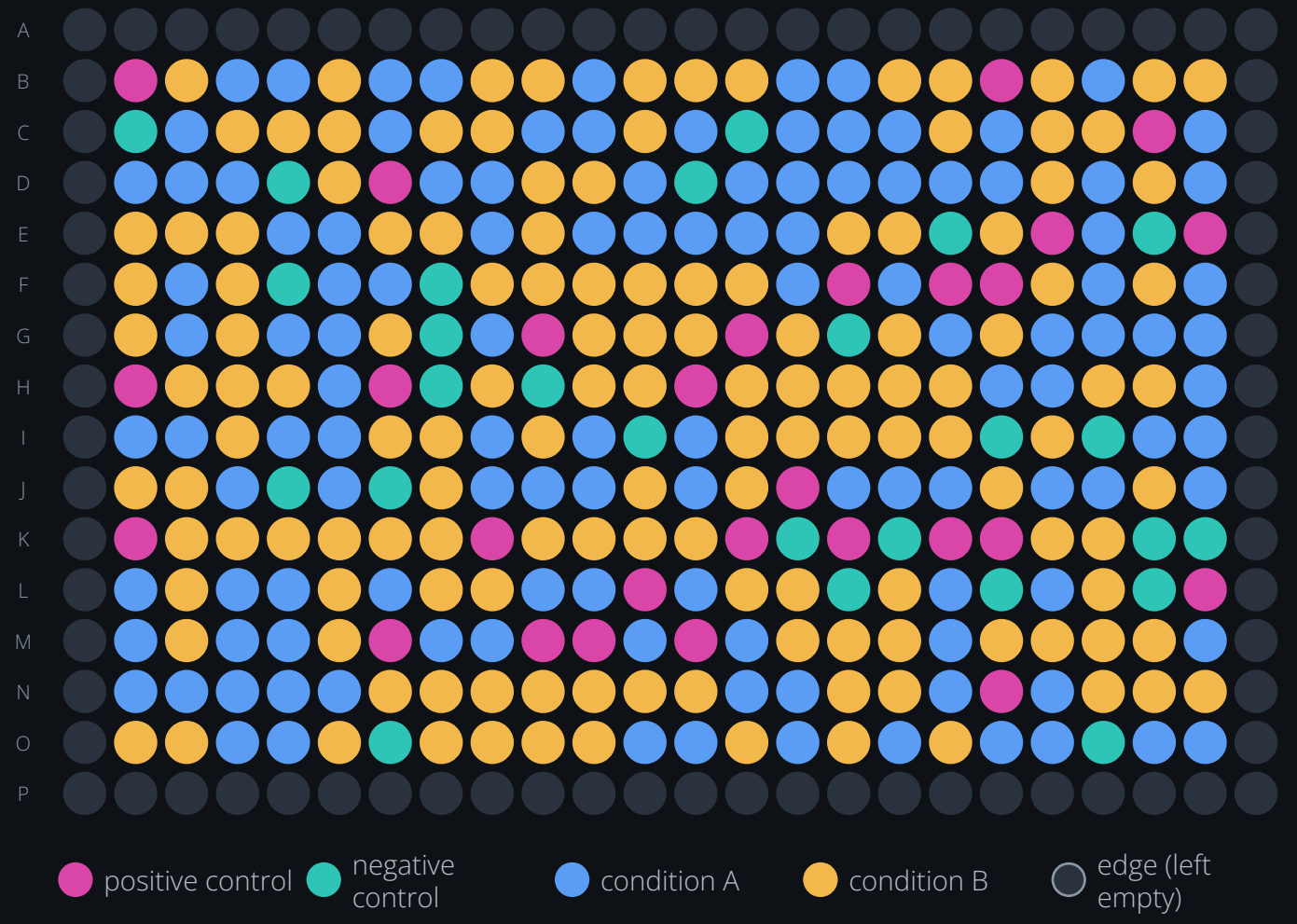


Illustration: a randomised 384-well layout with edge wells left empty

DATA · 5 OF 7

Power / Design: how big a screen?

BETA

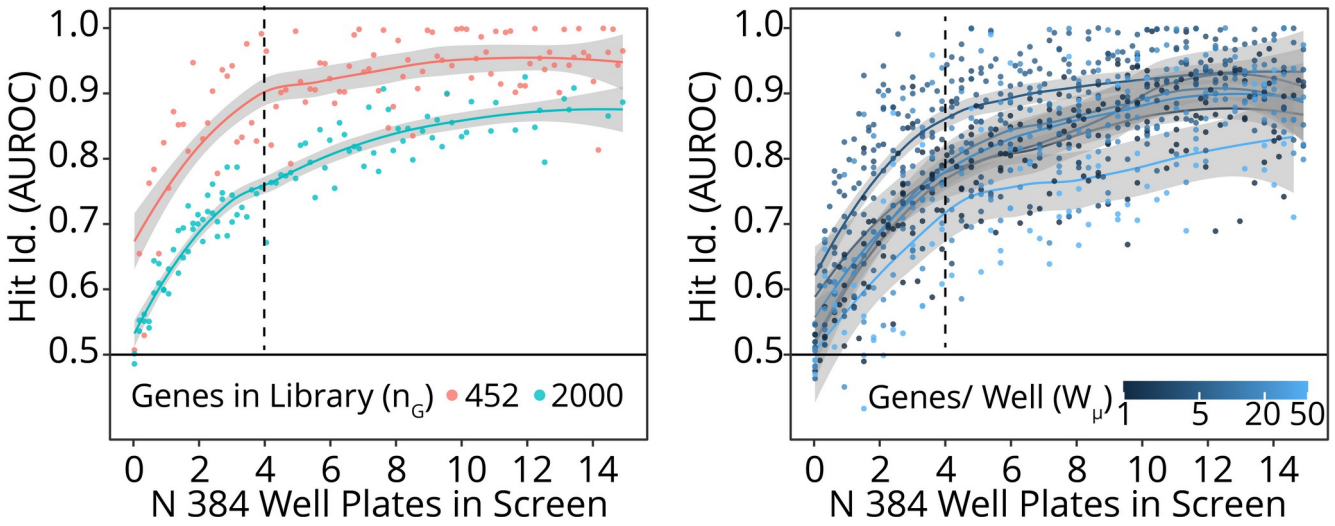
Estimates how many cells per well and how many plates you need to find a real hit.

WHAT YOU CAN DO

- Enter the library: genes, gRNAs per gene, constructs per well
- Enter the plate format and number of plates
- Enter the classifier's background rate and the effect you expect
- Simulates the screen many times to see how often the real hit is found

OUTPUT

- Detection probability vs cells per well
- A one-sentence answer, with caveats



spaCRPower simulations: hit detection vs plates, by library size and genes per well (preprint Fig. 1D–E)

DATA · 6 OF 7

Dose–Response: EC50 estimates

ALPHA

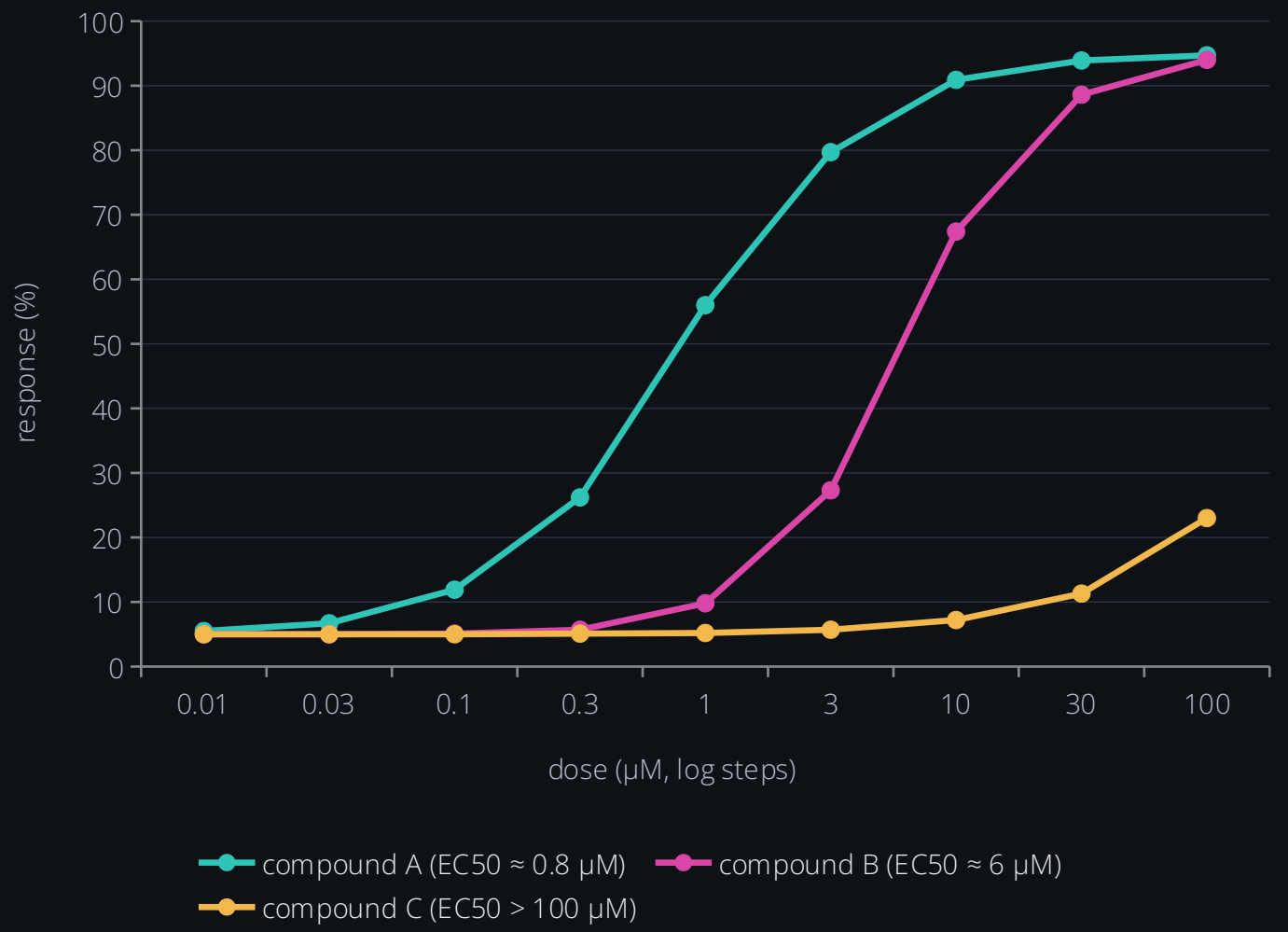
Fits dose–response curves per gene or compound, and says so when an EC50 can't be pinned down.

WHAT YOU CAN DO

- Choose concentration, response and grouping columns
- Fits the standard S-shaped curve
- A confidence interval on every EC50

OUTPUT

- Curves on a log-dose axis
- EC50 table with CI. Out-of-range → a bound ('EC50 > 30 µM'); failed fits listed as 'refused'
- Warns when a curve does not fit well



Schematic of the output (simulated curves)

DATA · 7 OF 7

QC: run quality checks

ALPHA

Five stored checks on one page. Reads what the run already wrote; recomputes nothing.

WHAT YOU CAN DO

- Segmentation QC scorecards
- Pixel size recorded
- Classifier tested on truly unseen wells
- Plate effects (row / column / edge)
- Annotator agreement

OUTPUT

- Verdict cards: ok · warn · fail · missing · stale
- Nothing is written to disk

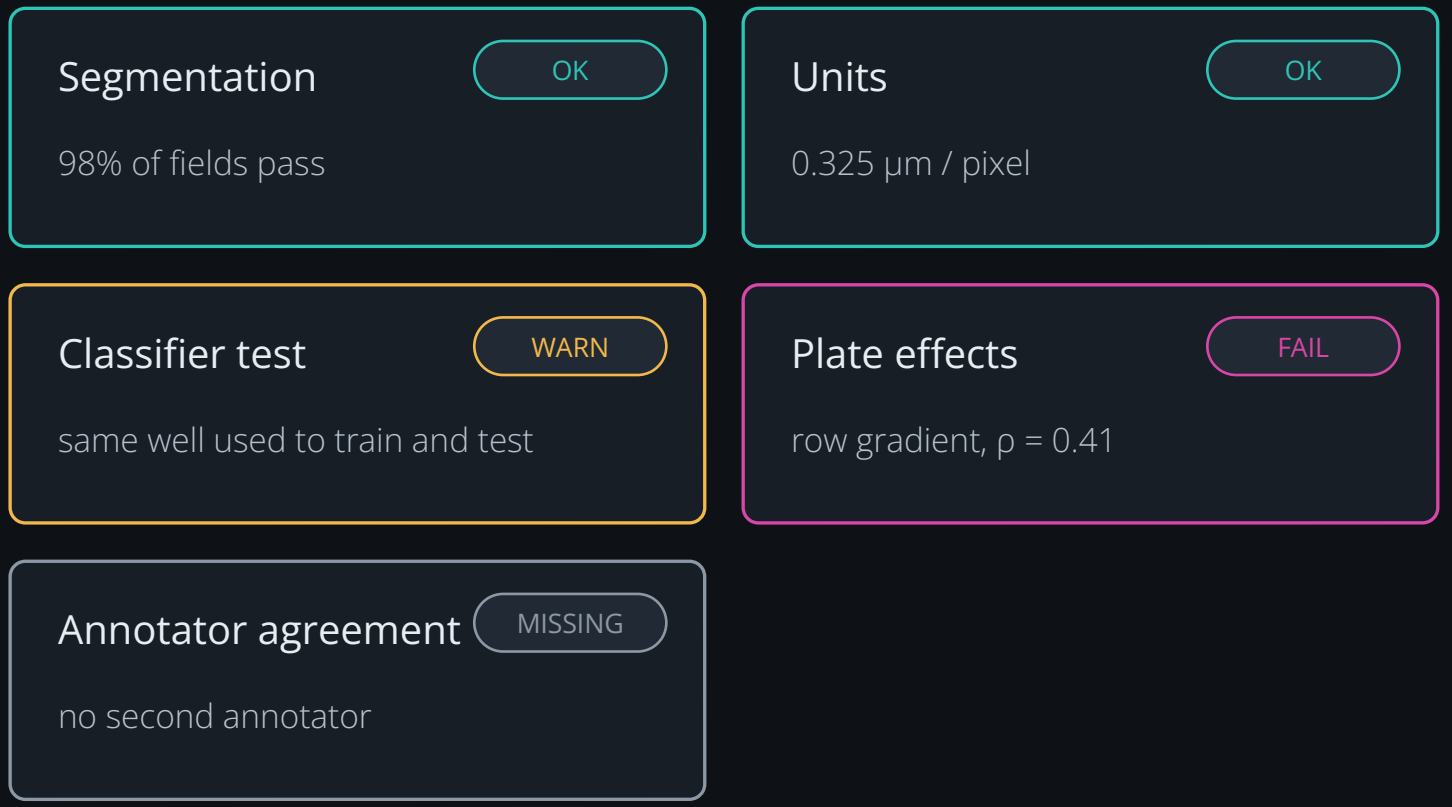


Illustration of the dashboard's verdict cards (values made up)

SUB-TOOLS INSIDE THIS MODULE

Layer Viewer
Images and masks as layers you can switch on and off

Control Charts
Follow a control over time and flag drift

Outliers
Find unusual cells or wells

03

Tools

Point these at a project: fix masks, stitch tiles, explore cells, gate populations, plot.

Make Masks

Align & Stitch

Image UMAP

Gate Editor

Graph Builder

TOOLS · 1 OF 5

Make Masks: fix masks by hand

BETA

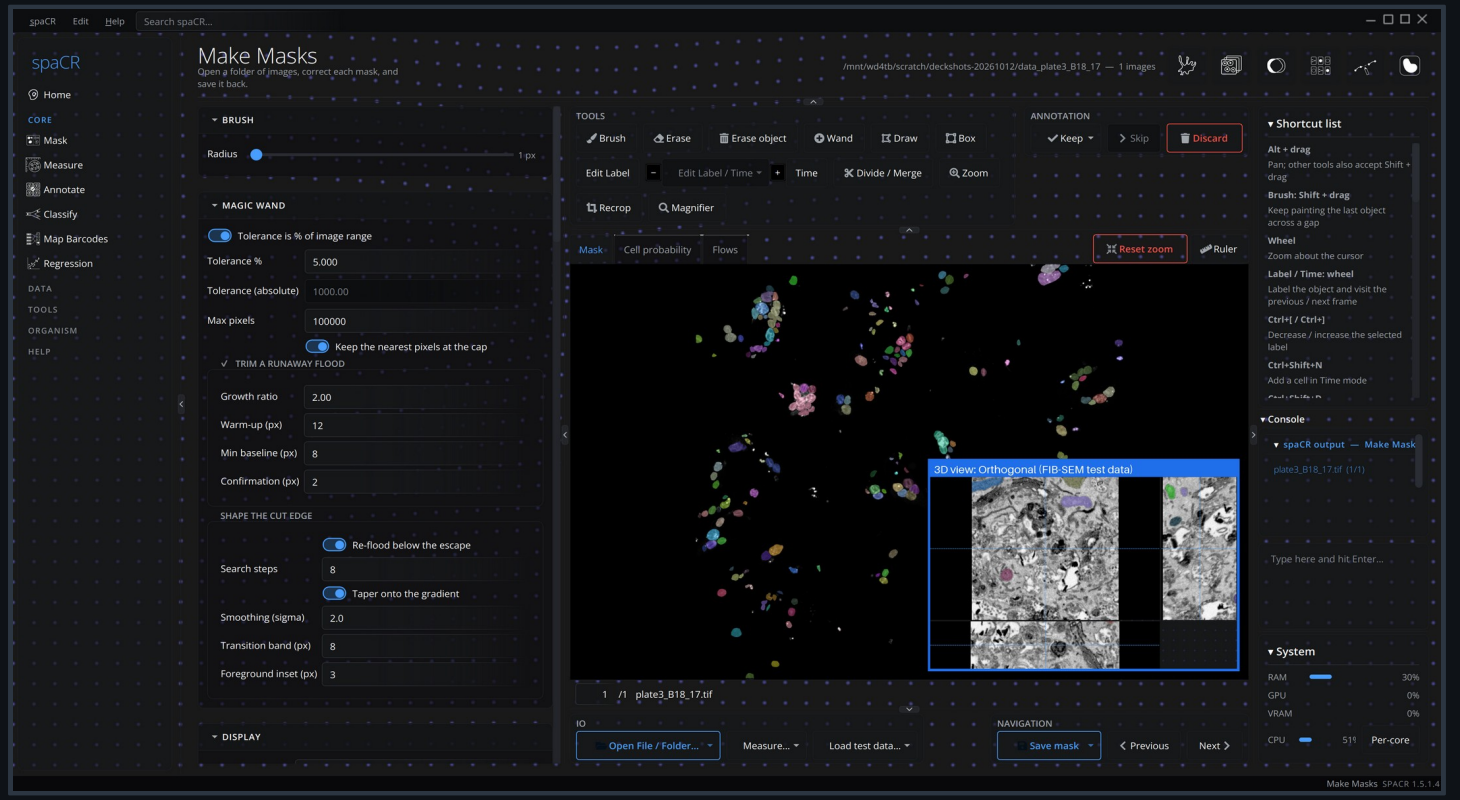
Correct segmentation by hand, and build training data for your own Cellpose model.

WHAT YOU CAN DO

- Tools: brush, erase, erase object, magic wand + / −, draw, box, divide, zoom, recrop, ruler
- Live magnifier: a box that zooms in and outlines cells as you move
- Edit Label and Time mode: object numbers kept across frames (32-bit labels); + Cell and + Daughter record divisions
- 3D in the same view: slice slider, Z wheel, View list; a 3D switch for object detection
- TEM images (DM3, DM4, MRC) with an Electron microscopy contrast preset; the pixel size is kept
- Open a folder or Load test data; Keep, Skip or Discard per field

OUTPUT

- Curated masks, ready for training or measuring
- A log of every edit, so curated masks are told apart from model output

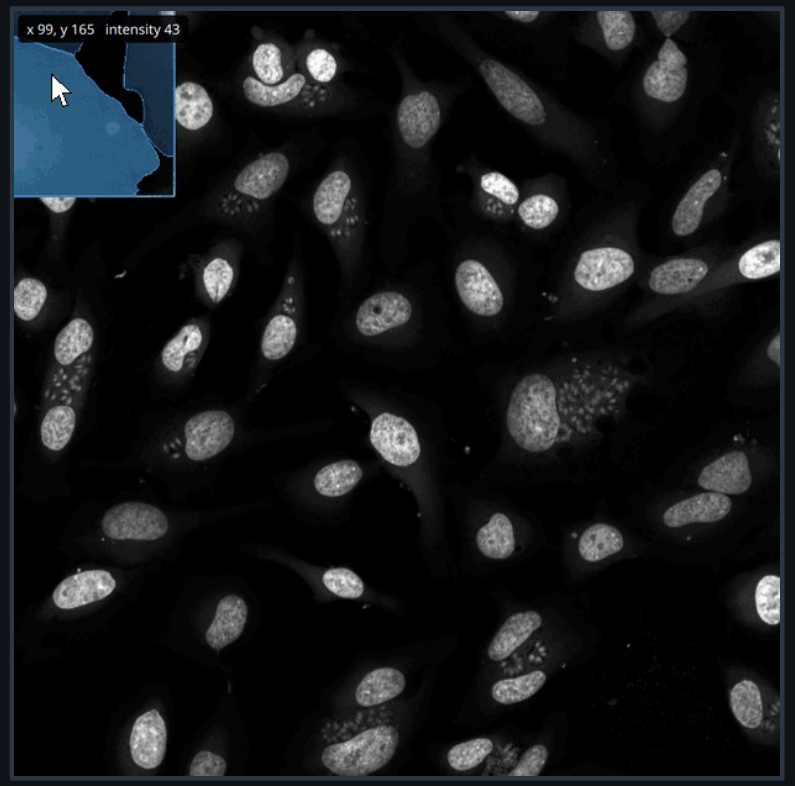


Make Masks: tools across the top, settings left, shortcuts right, field actions below. Inset: the Orthogonal 3D view on FIB-SEM test data

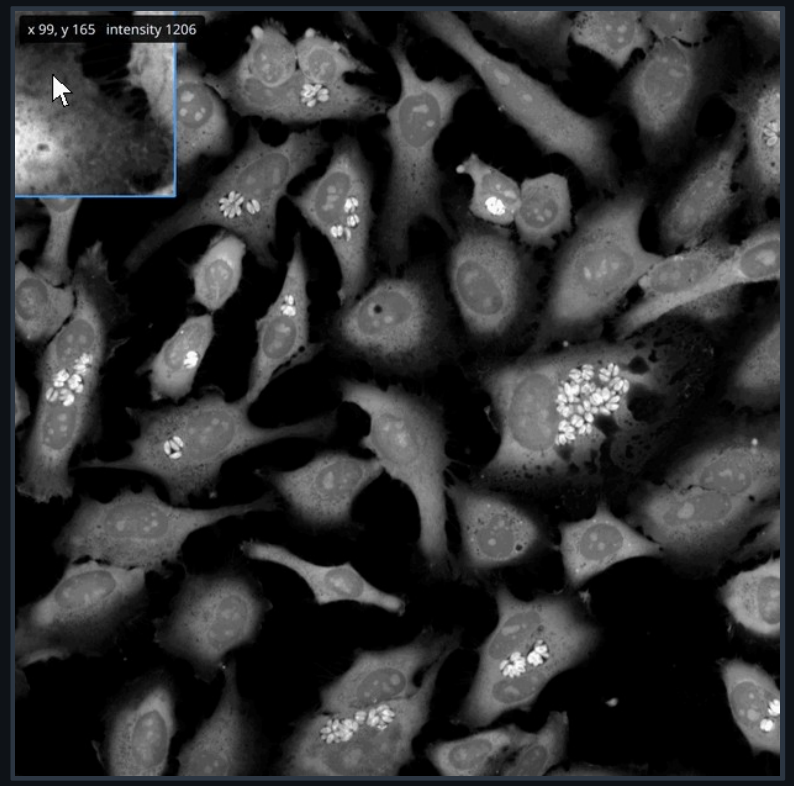
TOOLS · 1 OF 5

Make Masks: live magnifier

BETA



Cell from Hoechst: whole cells found from the nuclear (Hoechst) image



Toxoplasma from Cell Mask: vacuoles found from the cell-mask image

Box
follows the mouse, zoomed and contrast-adjusted

Model
Cellpose-SAM, your own model, or a threshold, outlining live

Click
adds what the box found; every click can be undone

Drag
along a cell bigger than the box joins its pieces into one object

TOOLS · 1 OF 5

Make Masks: Features

BETA

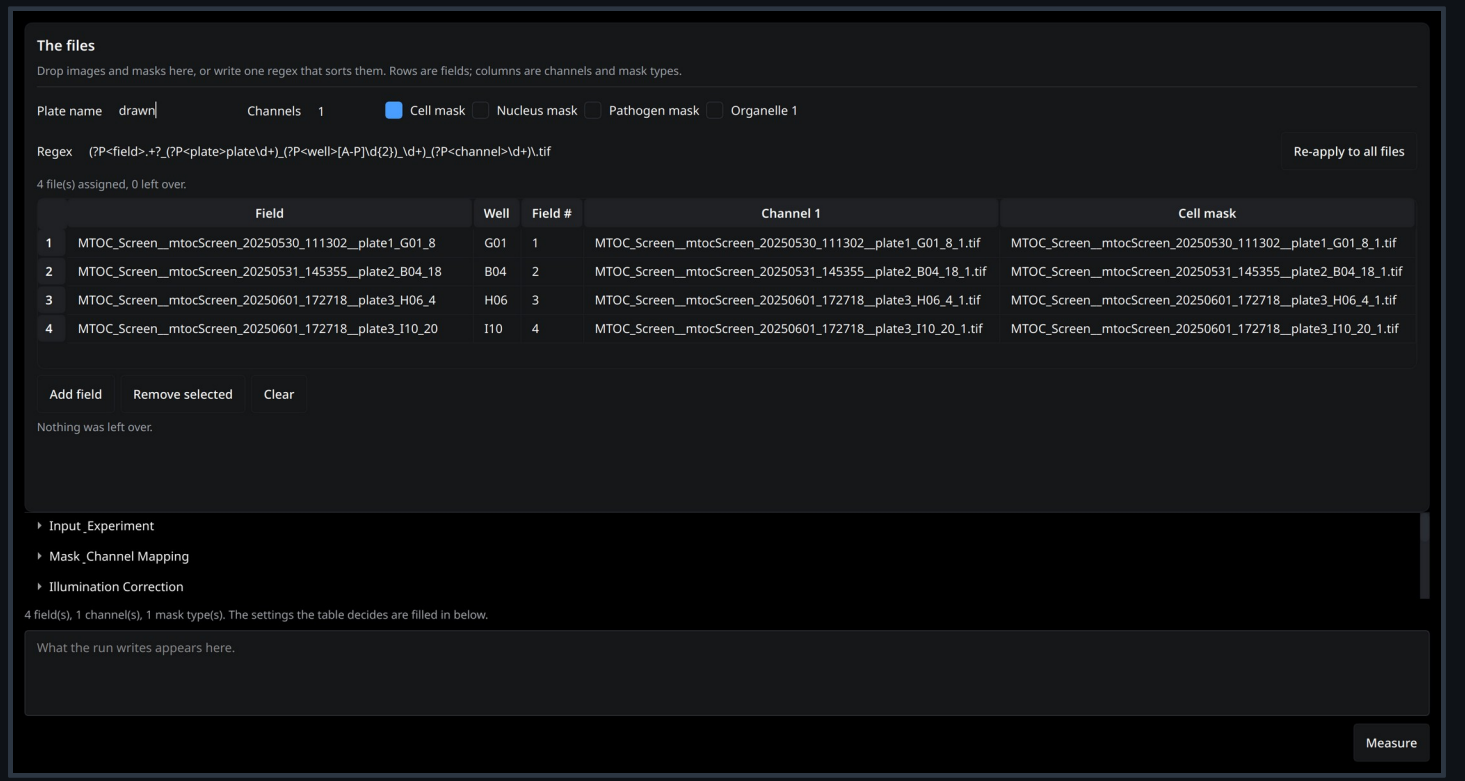
Measure the masks you drew, without leaving Make Masks.

WHAT YOU CAN DO

- One row per field; columns are the image channels and the mask types
- Fill it by dropping files, clicking a cell to pick a file, or one filename pattern
- Press Measure: the run goes through the Measure module itself

OUTPUT

- The same measurements database and crops a Measure run makes, from your hand-corrected masks

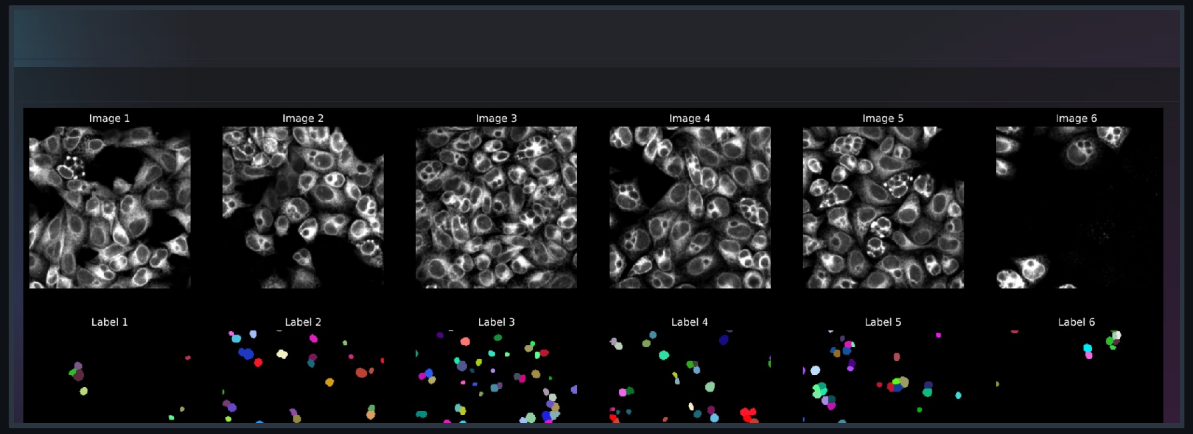


Features: four fields with their cell masks, ready to measure

TOOLS · 1 OF 5

Make Masks: train and compare models

BETA



Cellpose Workbench, training: image / label pairs it will learn from

Cellpose Workbench, applying a model: image, mask overlay and flows

SUB-TOOLS INSIDE THIS MODULE

Cellpose Workbench
Fine-tune Cellpose on your masks, then mask the whole folder

Model Compare
Two models' masks side by side, and how much they agree

Model Zoo
Browse, download and test trained models

Curate
Edit masks and tracks, with an edit log

Napari Bridge
Edit a mask in napari and bring it back

TOOLS · 2 OF 5

Align & Stitch: tiles to mosaic

STABLE

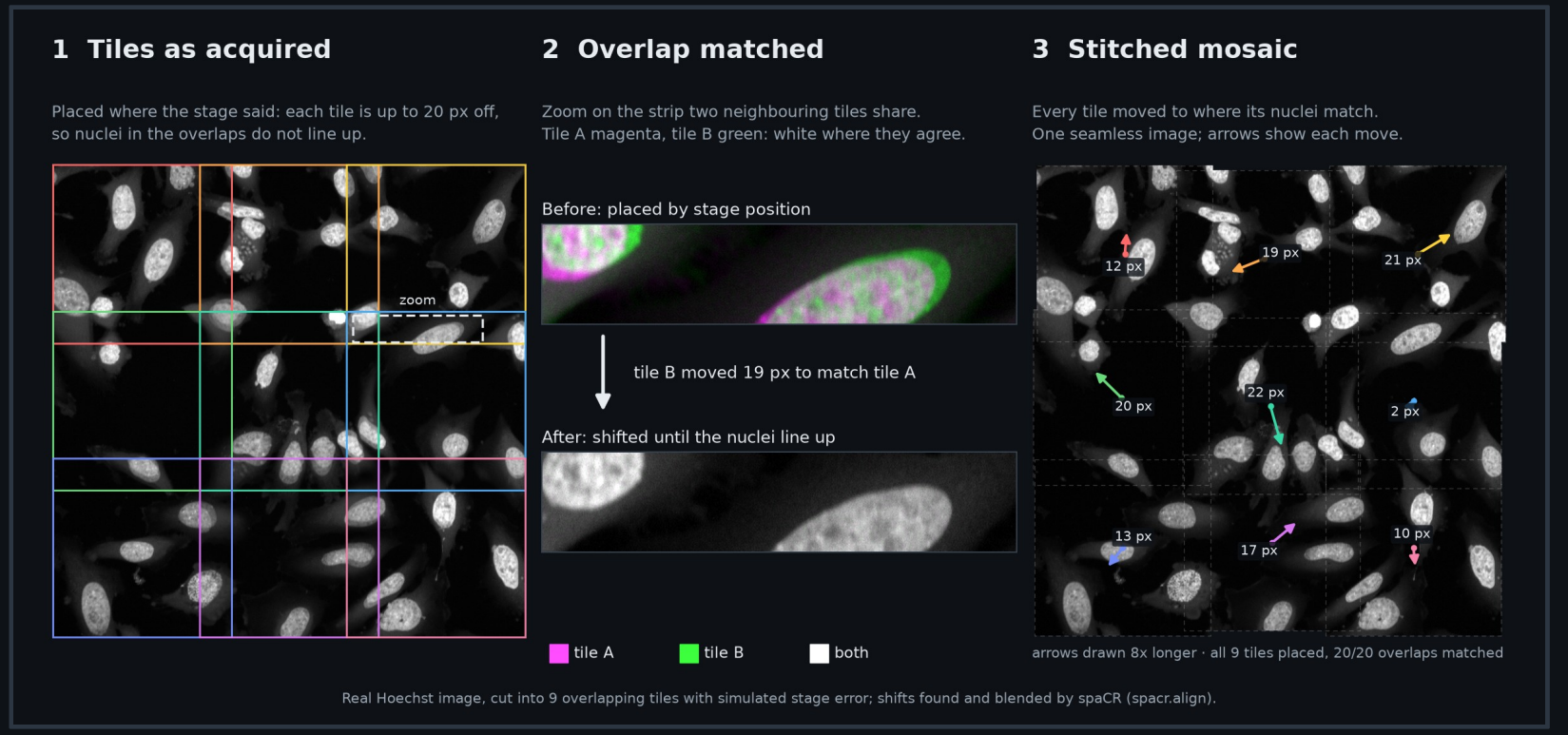
Stitches overlapping tiles into one large image without running out of memory.

WHAT YOU CAN DO

- Set grid, overlap, tile order and reference channel
- Plans first, reading only the overlaps: fast on huge plates
- Finds how far each tile is off by matching nuclei in the overlaps
- Shows how sure it is about every tile
- Tells you the final size and memory needed before it starts

OUTPUT

- The stitched mosaic, written piece by piece so memory stays low
- Where every tile ended up, saved with the project



A real Hoechst image cut into 9 overlapping tiles, each up to 20 px off (simulated); spaCR put every tile back where it came from

TOOLS · 3 OF 5

Image UMAP: map all cells

BETA

Places every cell on a 2-D map where similar cells sit together: with the real images.

WHAT YOU CAN DO

- UMAP, t-SNE or PCA maps
- Finds groups of similar cells automatically
- Embed by controls (positive vs negative groups)
- Can choose the map settings for you
- Click a point to see that cell's crop

OUTPUT

- Scatter coloured by cluster or condition, with thumbnails
- Table with a cluster column per cell

5,000 real cells from TSG101 screen plate 1: they fall into groups by number of nuclei and vacuoles

SUB-TOOLS INSIDE THIS MODULE

Image Scatter
Any two measured features, with the crop shown on hover

PCA
Which measurements drive the main differences

TOOLS · 4 OF 5

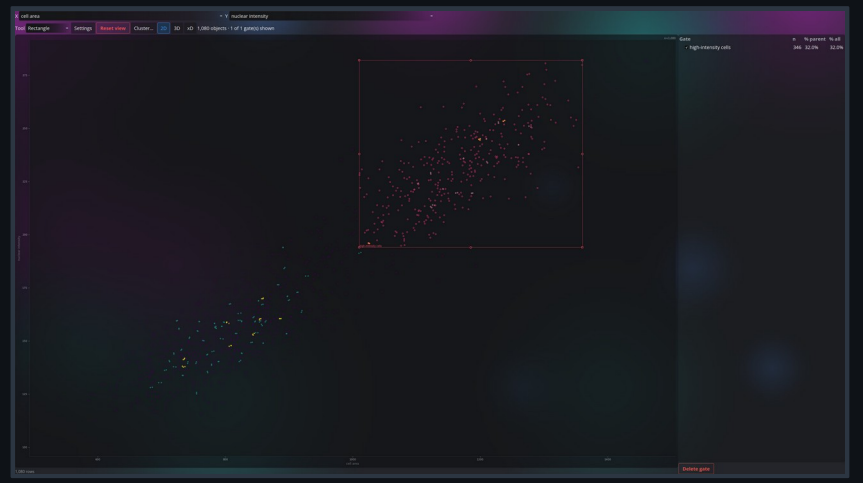
ALPHA

Gate Editor: gate cells like FACS

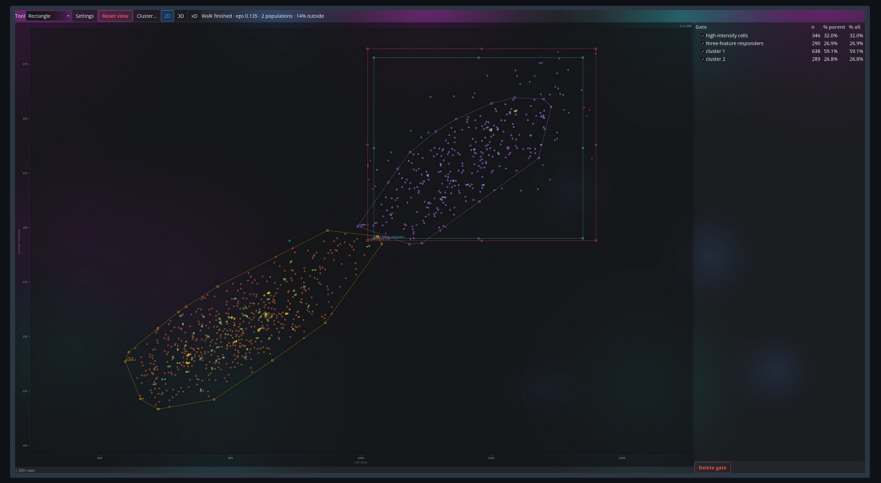
Select cell populations the way a flow cytometrist gates: in 1, 2 or 3 dimensions.

WHAT YOU CAN DO

- Load tables from one or several projects, or a CSV
- Threshold, rectangle, oval, polygon, wand gates in 1-D and 2-D
- Box, cylinder and prism gates in 3-D
- Nest and combine gates; count and % of parent for each
- Publish a gate as a filter every linked view follows
- Keys: Shift+E Edit, Shift+S Spin, Shift+D Draw, Shift+X / Y / Z for the plane (rebindable)
- Overlap picker: a click on stacked gates lists them all; choose which one to move



2-D rectangle gate on cell area vs nuclear intensity



Cluster walk: populations found and gated automatically

OUTPUT

- A gating strategy you can reuse on other plates
- Gate-membership columns in the project; PNG / PDF plots

TOOLS · 5 OF 5

Graph Builder: plots without code

BETA

Build a plot by dragging columns onto axes. The chart type follows the data.

WHAT YOU CAN DO

- Six drop zones: x, y, colour, size and two facets
- Histogram, bar, scatter, box or heatmap chosen for you; violin and line on request
- Brush points to select cells: linked to the UMAP and plate map
- Annotate conditions from metadata; reusable schemas
- Merge tables by name; save a chart and load it back

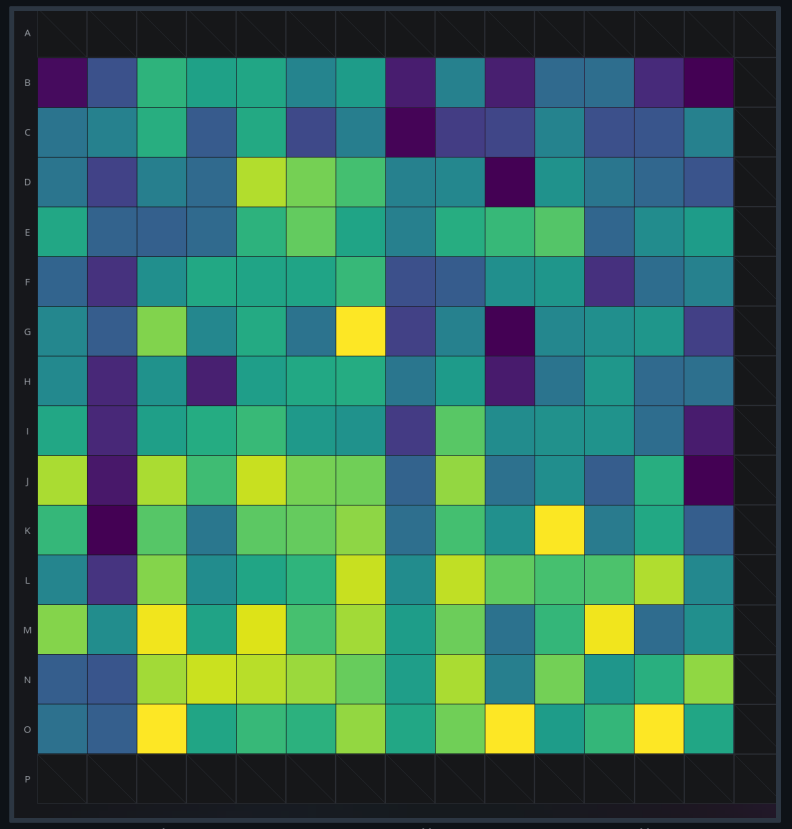
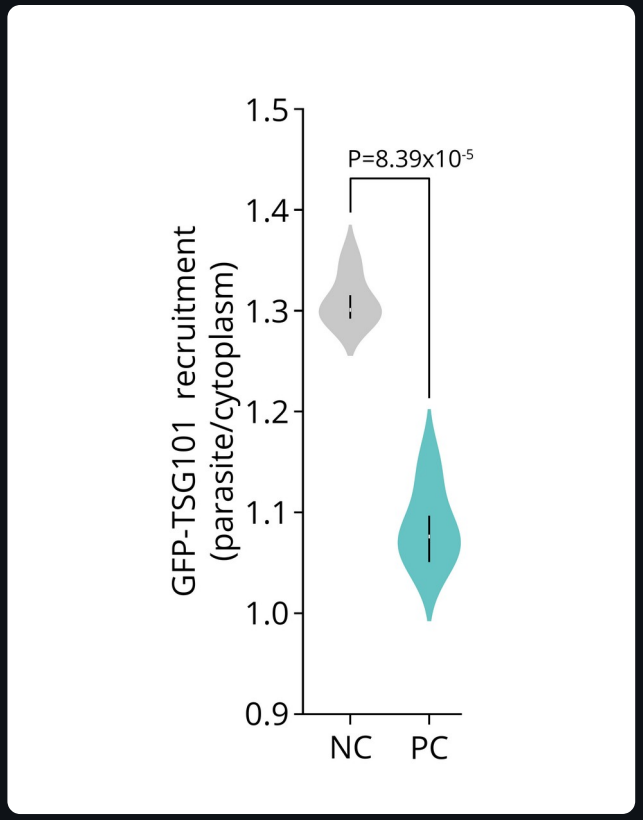


Plate Viewer: mean cell intensity per well



Violin plot of per-well values

OUTPUT

- Interactive charts and annotated tables, exportable

SUB-TOOLS INSIDE THIS MODULE

Plate Viewer
Plate heatmaps and edge-effect detection

Small Multiples
One plot per group, shared or independent axes

04

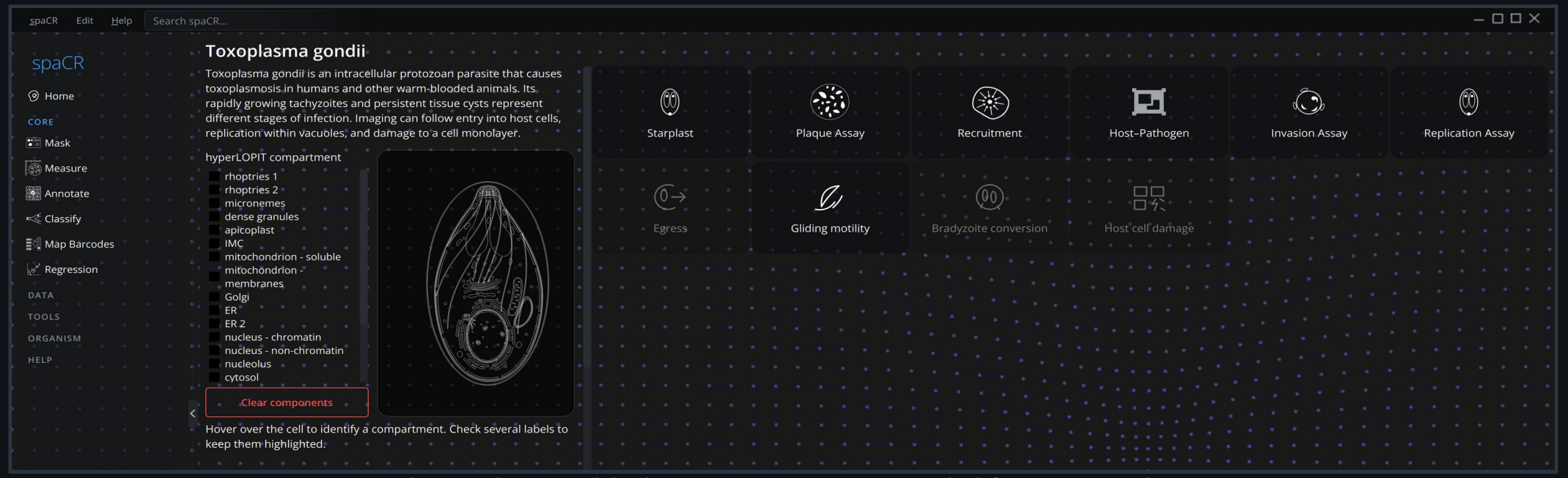
Assays

One module per organism, each with its own assays. Toxoplasma ships today; more organisms are alpha species.

Toxoplasma

ASSAYS

One module per organism



The Toxoplasma module: the parasite's compartments on the left, its assays as tiles

Toxoplasma

Plaque · Recruitment · Host–Pathogen · Invasion · Replication · Gliding motility, with ParaScape's hyperLOPIT compartments

More organisms (alpha)

Plasmodium, Candida, Trypanosoma, Leishmania, Giardia, virus and mammalian modules: Preferences > Show alpha species

Greyed tiles are assays still being built.

Plaque Assay: count and size plaques

BETA

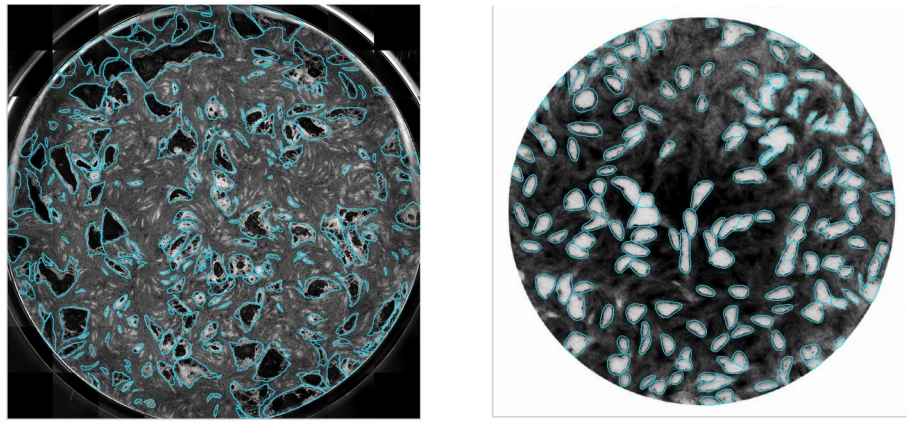
Finds every plaque in whole-well images and measures it in mm².

WHAT YOU CAN DO

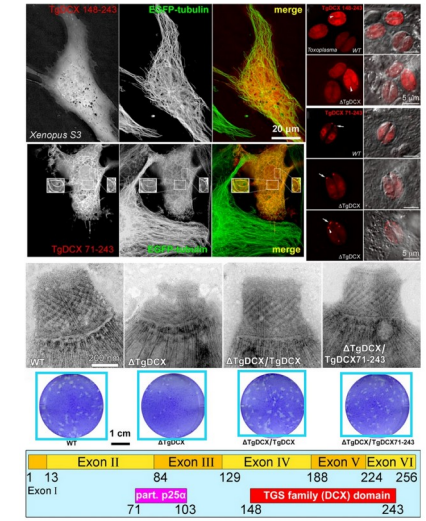
- Segments plaques with the Model Zoo plaque model (Cellpose-SAM)
- A YOLO detector finds and splits wells first
- Areas converted to mm² from the well diameter
- 'Figure' mode for plaque images taken from papers
- Can reuse existing masks

OUTPUT

- A table of every plaque, and a summary per well
- Area, perimeter, diameter, eccentricity, solidity per plaque



Curated plaque outlines: our microscope (left) and a published figure (right)



Wells found in a figure

ASSAYS · TOXOPLASMA · 2 OF 4

STABLE

Recruitment: marker at the vacuole

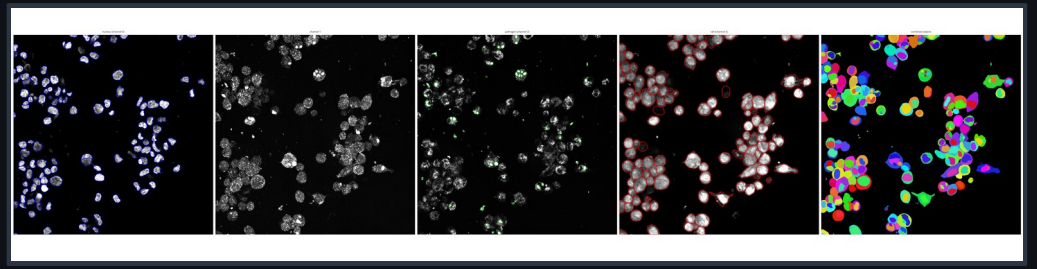
How much does a fluorescent marker gather at the parasite vacuole compared with the host cytoplasm?

WHAT YOU CAN DO

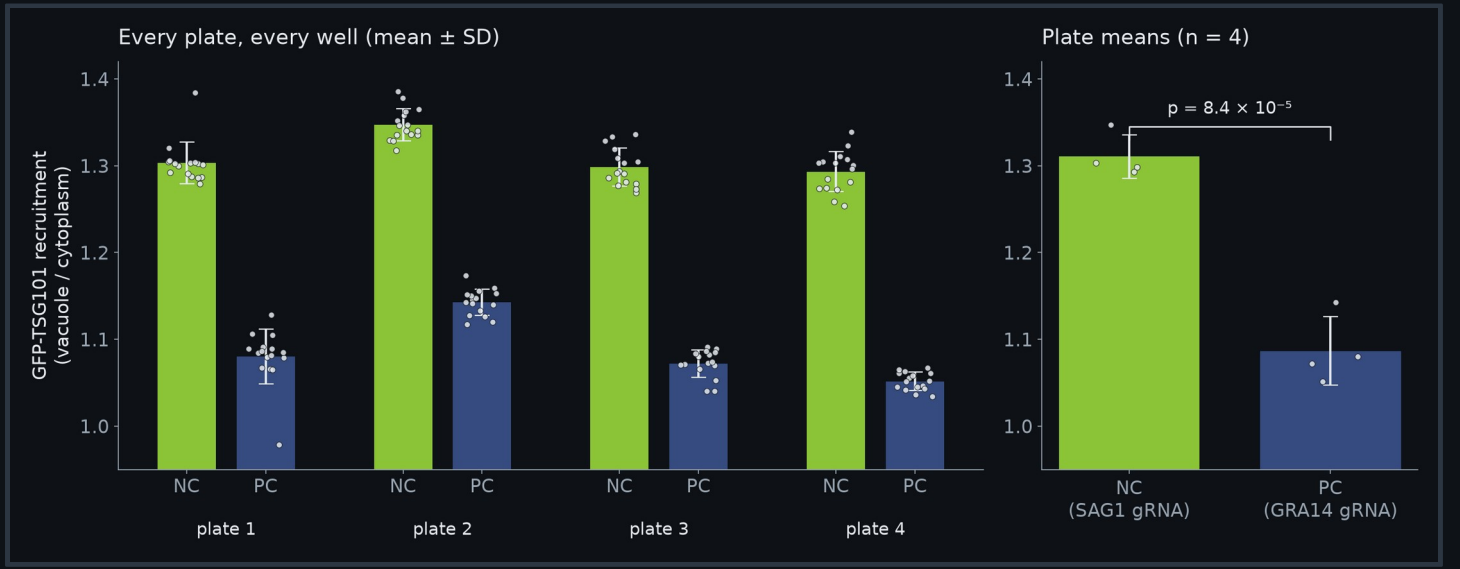
- Map wells to cell type, pathogen and treatment
- Size and intensity filters per object class
- Choose the channel of interest
- Minimum number of cells per well

OUTPUT

- Tables per cell and per well
- Bar plots of recruitment by condition, with per-well points



One field: channels and the cell / nucleus / parasite masks



GFP-TSG101 recruitment to the vacuole, NC vs PC, 4 plates (preprint data)

ASSAYS · TOXOPLASMA · 3 OF 4

STABLE

Invasion: attached or invaded?

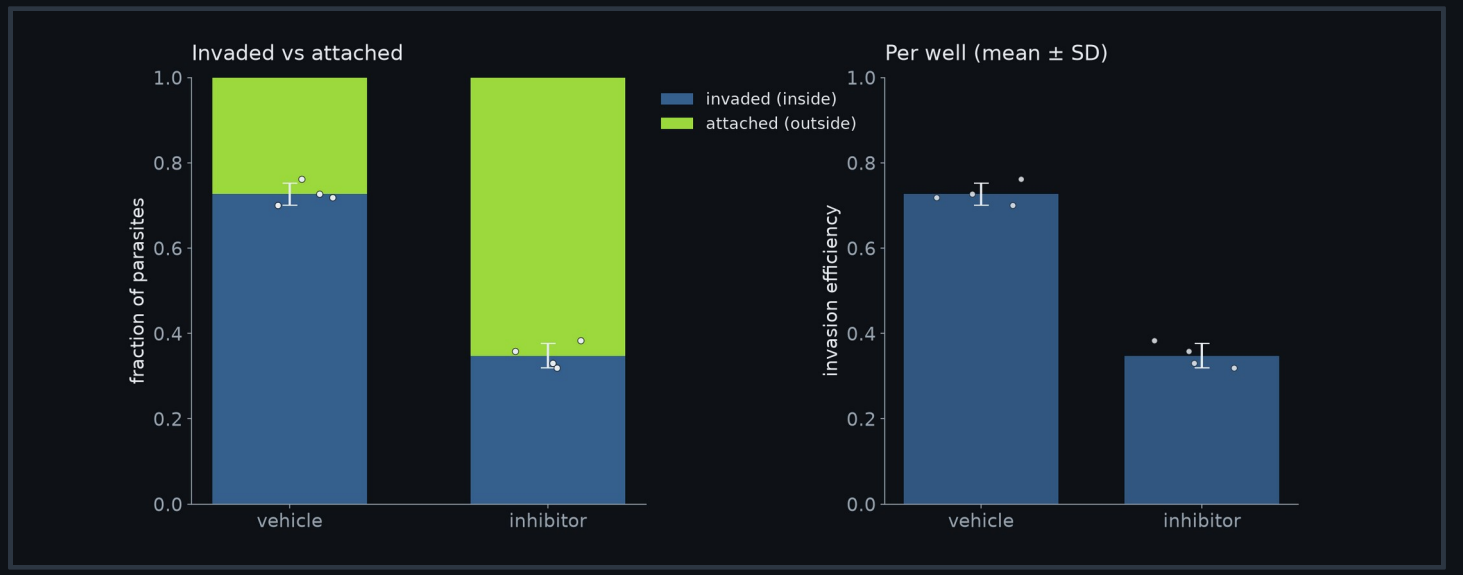
Scores each parasite as outside (attached) or inside (invaded) from two-colour differential staining.

WHAT YOU CAN DO

- Set the outside-stain and total-stain channels
- Thresholds per field, or from control wells
- Flags fields where inside and outside cannot be told apart
- Checks the answer holds if the threshold moves

OUTPUT

- Per-parasite calls, per-field thresholds, per-well invasion
- Statistics comparing conditions, per well
- Plots per well, by condition, and of the thresholds



spaCR invasion analysis of the test dataset (4 wells per condition, mean ± SD)

Replication: parasites per vacuole

BETA

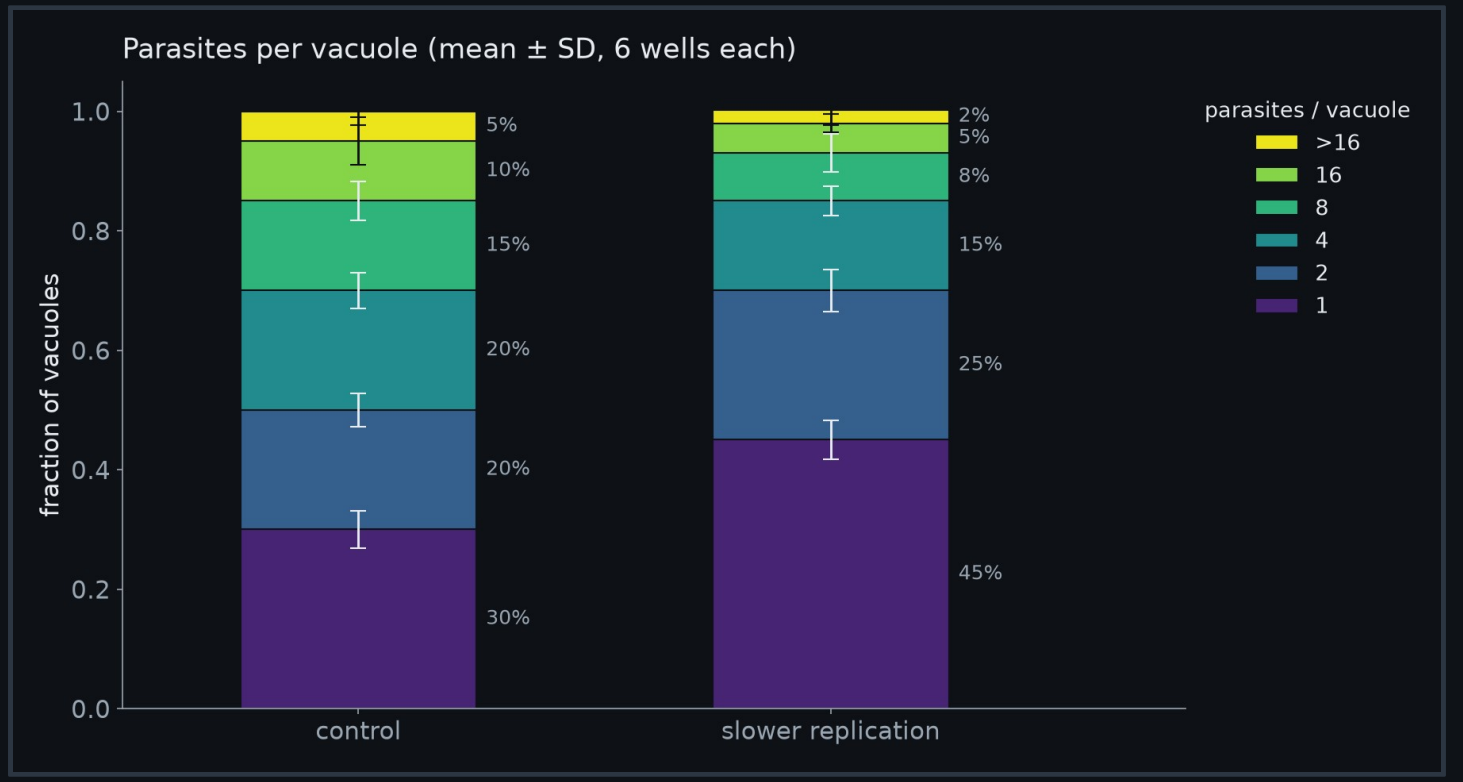
Counts parasites per vacuole (1, 2, 4, 8, 16 …) and compares the distributions between conditions.

WHAT YOU CAN DO

- Groups parasites into vacuoles: auto, spatial, by cell, or by object
- Area filters for parasites and vacuoles
- Warns when too many vacuoles have a non-power-of-two count

OUTPUT

- Vacuole counts, per-well distributions, condition summary
- Statistics comparing conditions
- Stacked-bar plots per well and by condition



Example output on synthetic data: 6 wells per condition, mean ± SD

In progress: a classification model that finds and counts Toxoplasma from the Toxoplasma stain, from the nuclear stain, or from the cell mask, so parasites per vacuole can be scored without a dedicated parasite channel.

BEHIND THE TILES

Project tools in the Help menu

Report
HTML / PDF report of a run: figures, statistics, settings

Batch Runner
Queue modules with dependencies; validate, then run unattended

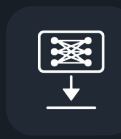
Run History
Every run, its settings and its outputs

Database Browser
Browse, search, filter and export project tables

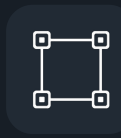
Project Browser
All projects on this machine

Pipeline Graph
Which run produced which file, as a graph

Plate Queue
Line up plates for Mask / Measure and let them run overnight



Distributed Jobs
Send work to SSH hosts, Slurm or the cloud



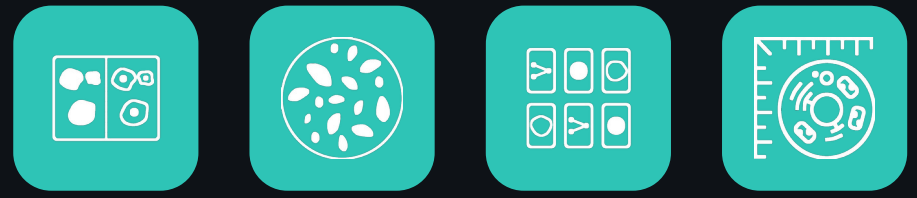
Data Manager
What is on disk, how big, and what can go

Feature Dictionary
What every measured column means, and its unit

05

Trained models

Models trained for Toxoplasma and host-cell images, shipped through the spaCR Model Zoo. Newest version of each.



MODEL ZOO

Eight models, one line in a settings file

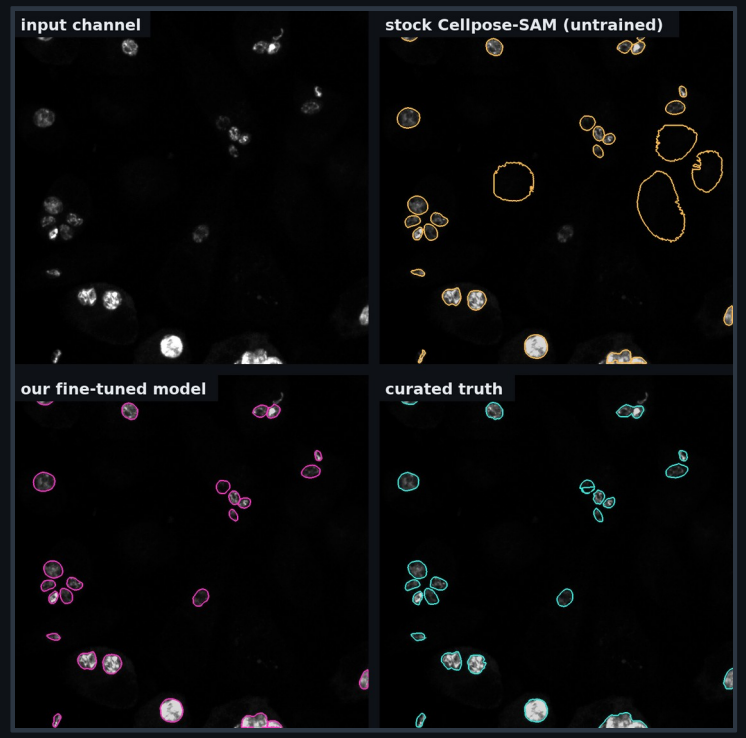
Model	Settings key	Finds	Architecture	Score on unseen images
Toxoplasma PV v4	`toxoplasma_pv_v4`	Parasite vacuoles	Cellpose-SAM	F1 0.761 (stock 0.377)
Toxoplasma Plaque v5	`toxoplasma_plaque_v5`	Plaques in wells	Cellpose-SAM	F1 0.816 (stock 0.227)
Plaque Well Detector v3	`toxoplasma_well_detector_v3`	Wells in plates / figures	YOLO11n	mAP50 0.994
Toxoplasma from Cell Mask	`toxoplasma_from_cellmask_v1`	PVs from the cell channel	Cellpose-SAM	F1 0.606 (stock 0.021)
Nuclei from Cell Mask	`nuclei_from_cellmask_v1`	Nuclei from the cell channel	Cellpose-SAM	F1 0.888 (stock 0.201)
Cell from Hoechst	`cell_from_hoechst_v1`	Cells from the nuclear channel	Cellpose-SAM	F1 0.870 (stock 0.301)
Toxoplasma from Hoechst	`toxoplasma_from_hoechst_v1`	PVs from the nuclear channel	Cellpose-SAM	F1 0.569 (stock 0.002)
Live cell v1	`live_cell_v1`	Unstained cells (phase, DIC)	Cellpose-SAM	F1 0.960 (stock 0.885) *

** Live cell v1: F1 0.960 vs 0.885 on data stock never saw; 0.694 vs 0.738 over all 2,199 test fields (stock trained on LIVECell).*

How it works Name a key in a settings file: `pathogen_model: toxoplasma_pv_v4`: and spaCR downloads it the first time it is needed. Anyone can contribute a trained model: it is uploaded to the spaCR Hugging Face and then appears in every user's Model Zoo.

MODEL · TOXOPLASMA_PV_V4

Toxoplasma PV v4



Unseen test field. Orange = vanilla Cellpose-SAM (cpsam_v2, never trained on our images), magenta = PV v4, cyan = curated truth

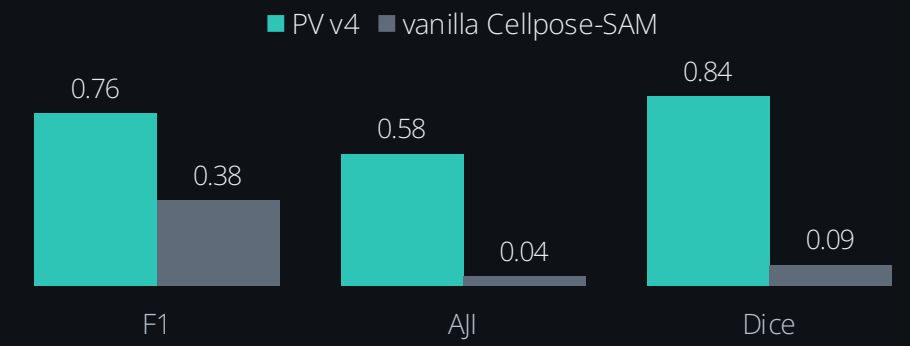
WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), round 7

Outlines Toxoplasma parasitophorous vacuoles in anti-Toxoplasma or DsRed images (RH and ME49).

209 held-out fields, never seen in training



TRAINED ON

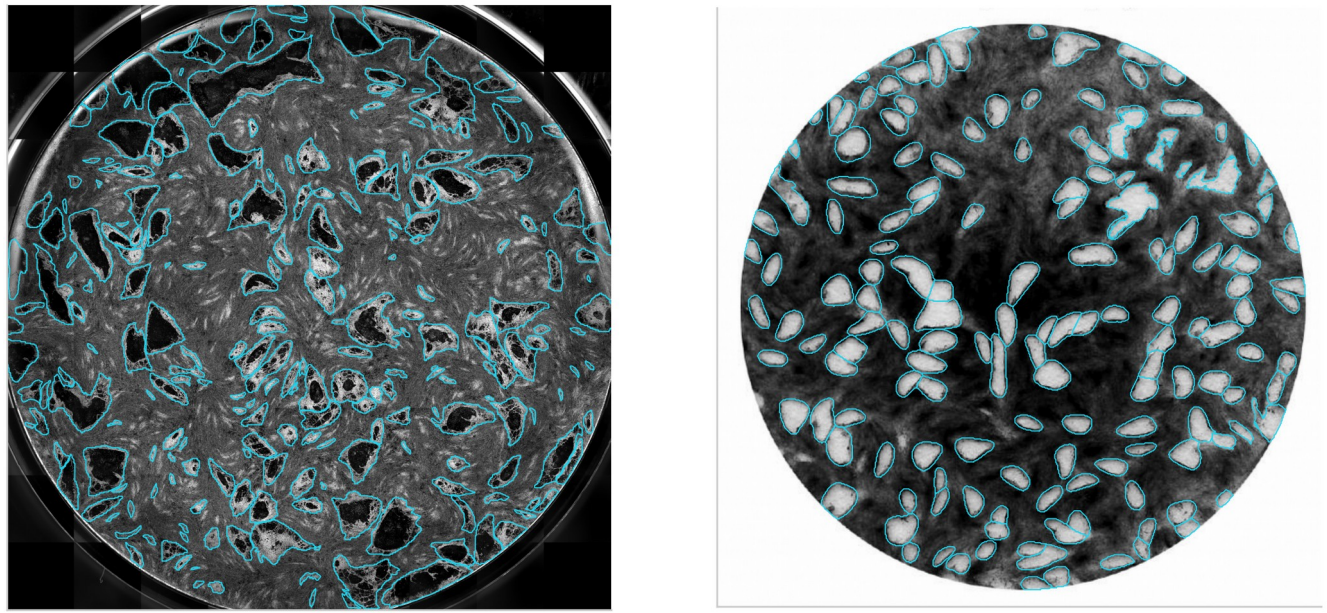
Round 6's 556 curated fields plus 80 from a new plate: 23,340 vacuoles in 502 training fields. 11 wells have never been used for training.

GOOD TO KNOW

- Re-scored on curated ground truth: F1 0.761, precision 0.77, recall 0.75
- On the new plate it was trained on, PV v3 finds more vacuoles (F1 0.90 vs 0.75): use v3 there
- Weak on published-figure crops, as is every PV model

MODEL · TOXOPLASMA_PLAQUE_V5

Toxoplasma Plaque v5



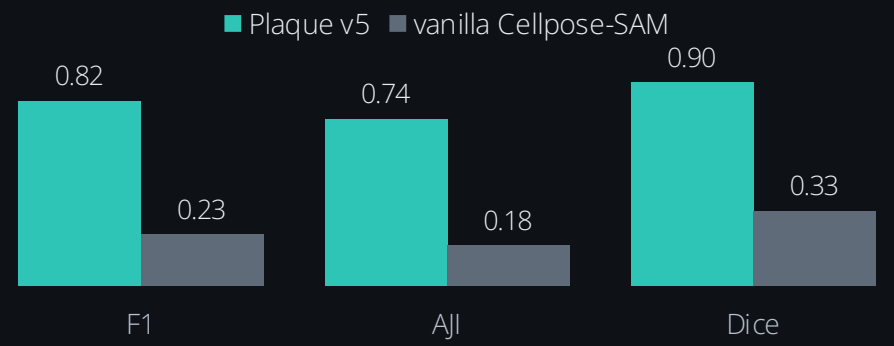
Curated plaque outlines (cyan): our microscope (left) and a published figure (right)

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), round 7
Outlines plaques in crystal-violet wells: from gel imagers, our plates and published figures.

153 held-out fields, never seen in training



TRAINED ON

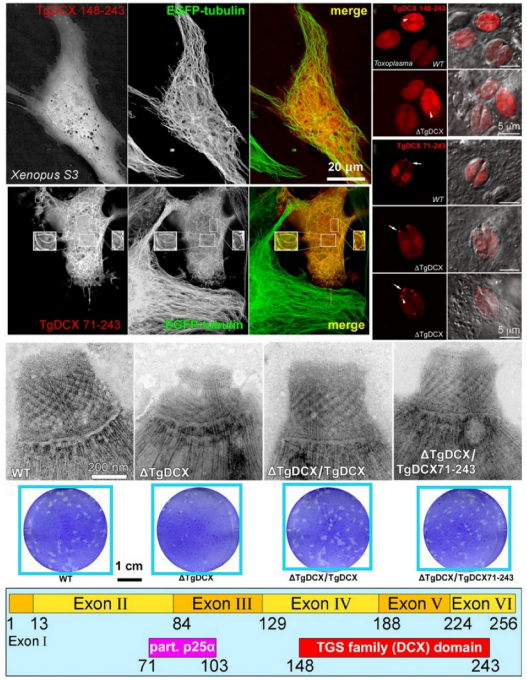
1,507 curated fields (1,201 train, 153 validation, 153 test): Gel Doc wells, lab plates and published figures; 49,360 training plaques.

GOOD TO KNOW

- Default since 2026-10-17: built on cpsam_v2
- Precision 0.87, recall 0.77; on the 153 test fields v4 scores F1 0.725 and v3 0.741
- On Gel Doc wells v3 scores higher (0.954 vs 0.868): use v3 there
- v4 stays selectable

MODEL · TOXOPLASMA_WELL_DETECTOR_V3

Plaque Well Detector v3



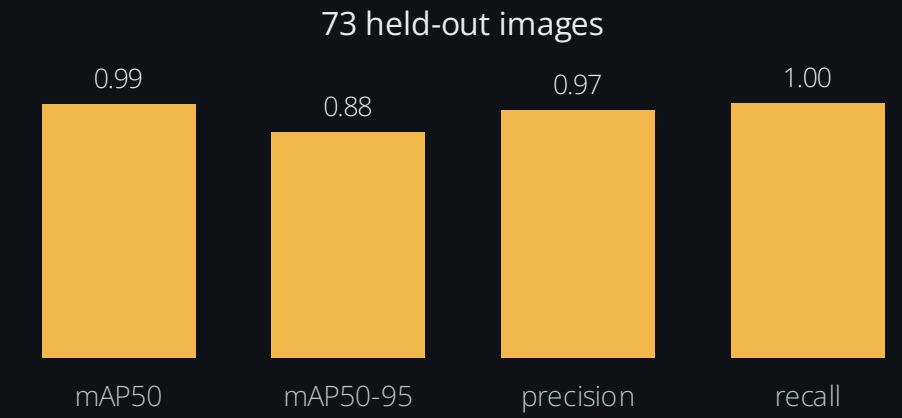
Detected wells (cyan) in a published figure; fluorescence and EM panels ignored. Shown with an earlier detector

WHAT THE SCORES MEAN

precision of the boxes it draws, the share that really are wells.
recall of the real wells, the share it finds.
mAP50 overall detection score; a box counts when it overlaps the true well by at least half.
mAP50-95 the same, but also rewards boxes drawn tightly around the well.

YOLO11n object detector (the default)

Draws a box around every plaque well in a plate photo or a published figure: the first step before plaque segmentation.



TRAINED ON

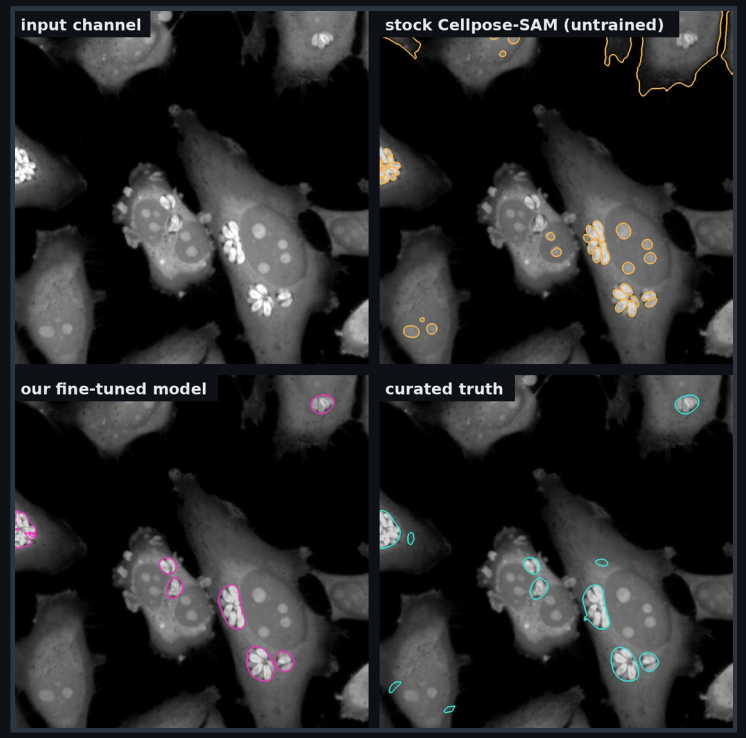
452 reviewed training images (1,679 wells): our plates, Gel Doc plates and figures from published papers, grouped by plate and figure.

GOOD TO KNOW

- The default well detector since 2026-10-12
- Its tight-box score (mAP50-95) is below the previous default's on the 73 clean images, and above it on 3 Gel Doc plates
- Finds wells, not plaques; the well size also gives the mm² scale

MODEL · TOXOPLASMA_FROM_CELLMASK_V1

Toxoplasma from Cell Mask



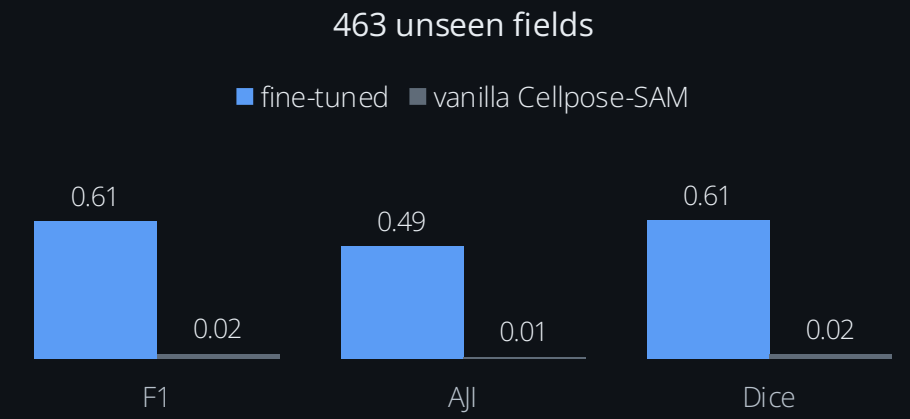
Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds Toxoplasma vacuoles from the host cell-mask channel alone: no parasite stain needed.



TRAINED ON

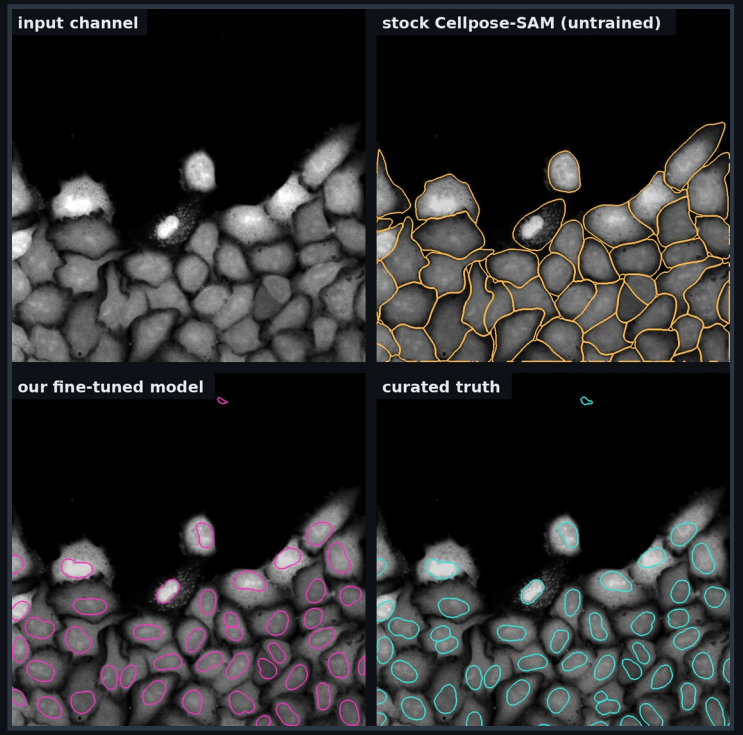
2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

GOOD TO KNOW

- Stock Cellpose cannot do this at all
- F1 by host: HeLa 0.71 · HFF 0.56 · THP-1 0.47
- For counting, occupancy and area: not fine morphology

MODEL · NUCLEI_FROM_CELLMASK_V1

Nuclei from Cell Mask



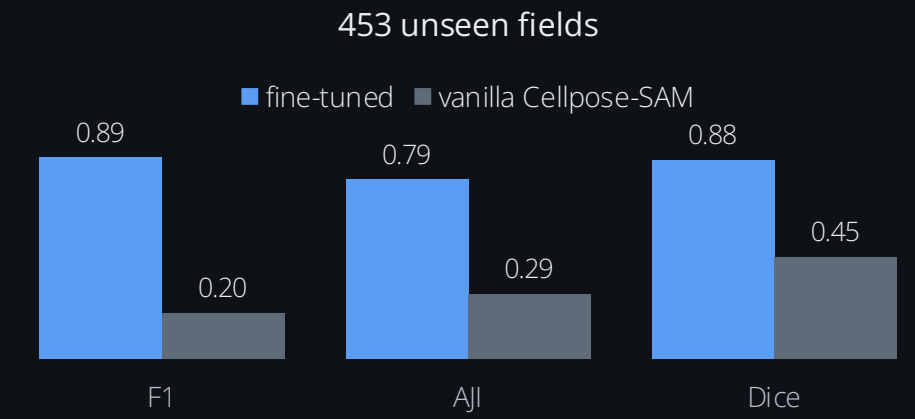
Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds nuclei from the cell-mask channel alone: frees the DAPI / Hoechst channel for another marker.



TRAINED ON

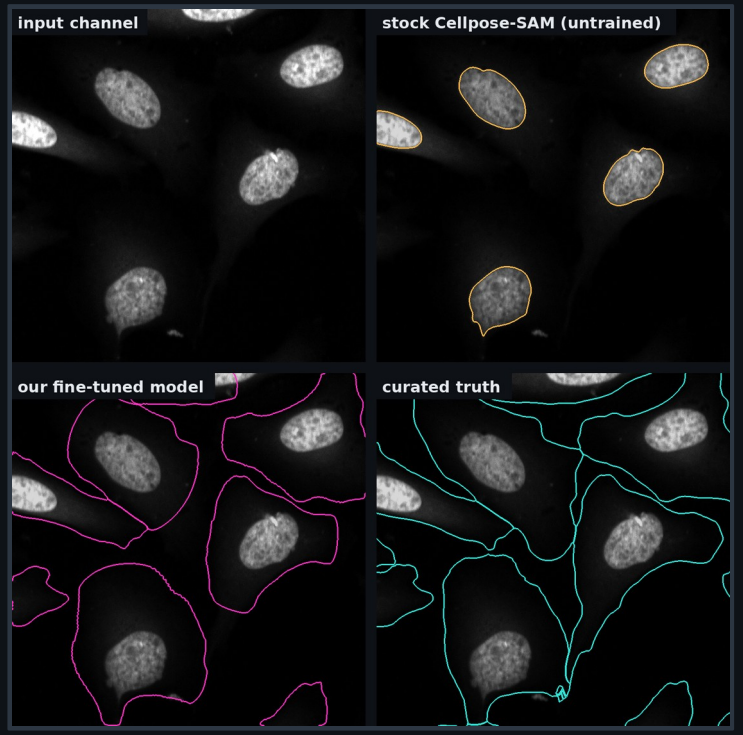
Cell-mask images paired with curated nuclear masks; tested on 453 fields from wells never used for training. HFF, HeLa and THP-1.

GOOD TO KNOW

- F1 by host: HFF 0.93 · HeLa 0.86 · THP-1 0.86
- Learns nuclei from cell shape: expect worse on unusual or very confluent cells

MODEL · CELL_FROM_HOECHST_V1

Cell from Hoechst



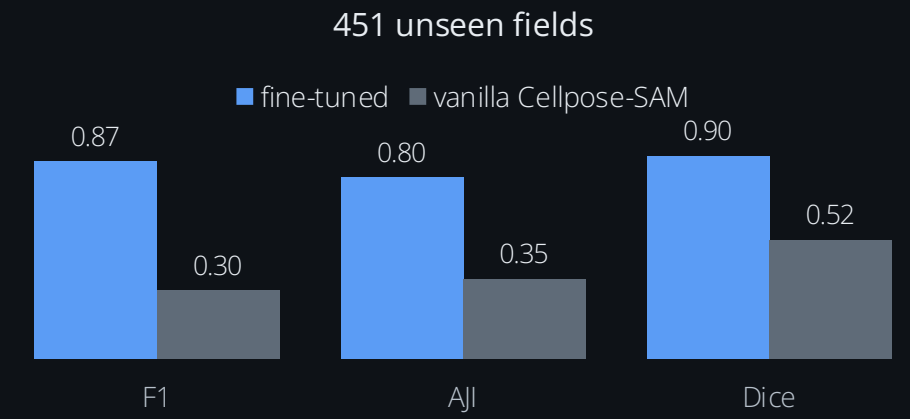
Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds whole host cells from the Hoechst (nuclear) channel alone: the counterpart of nuclei-from-cell-mask.



TRAINED ON

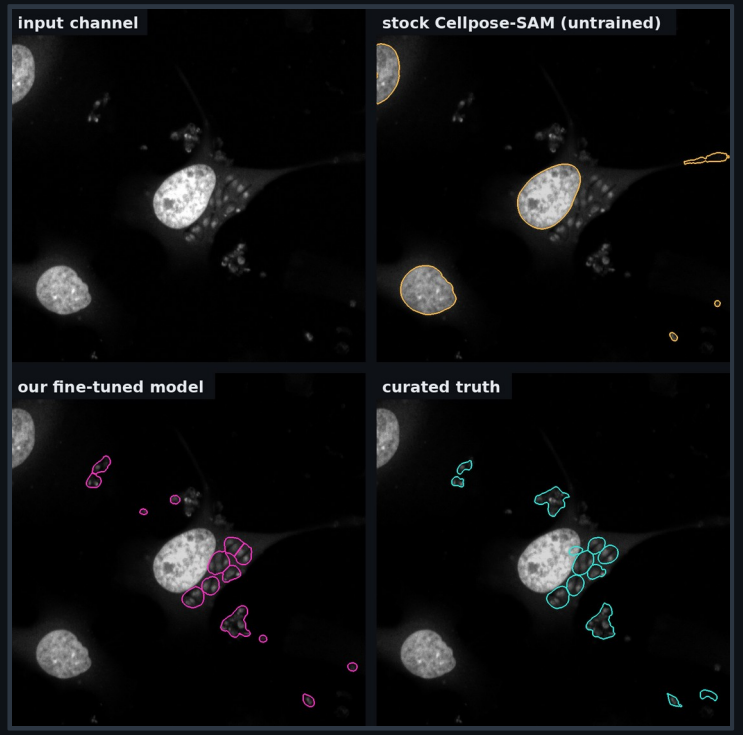
2,578 training fields (237,957 cells); tested on 451 fields (45,098 cells) from other wells.

GOOD TO KNOW

- Misses some cells but rarely invents one (precision 0.94, recall 0.81)
- F1 0.57 higher than stock

MODEL · TOXOPLASMA_FROM_HOECHST_V1

Toxoplasma from Hoechst

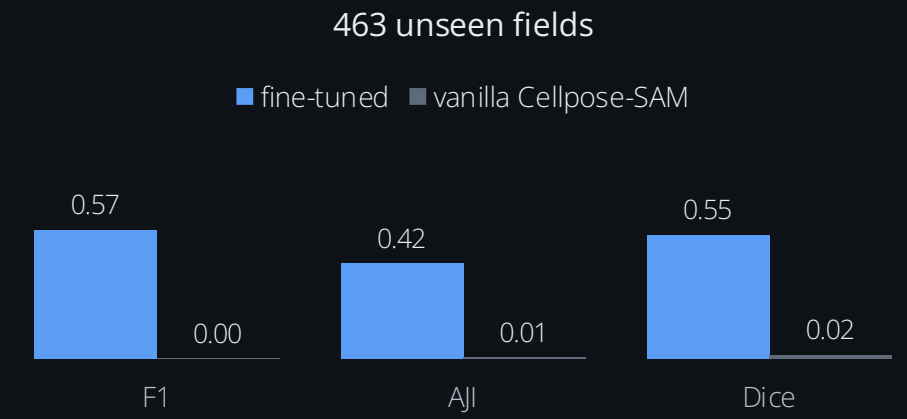


Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.
AJI how well each outline matches; lower if a cell is split in two or two cells merged.
Dice pixel overlap between the model's mask and the true mask.
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel
Finds Toxoplasma vacuoles from the Hoechst channel alone: no parasite stain needed.



TRAINED ON

2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

GOOD TO KNOW

- Harder than the cell-mask route (F1 0.57 vs 0.61)
- Stock Cellpose finds essentially nothing
- For counting, occupancy and area

MODEL ZOO

Other models available in spaCR

Segmentation

- Cellpose 4: Cellpose-SAM (default) and Cellpose-DINO
- Cellpose 3 models (cyto3, nuclei, ...)
- Cellpose models from bioimage.io
- Models other users have contributed

Image classifiers (Classify)

- ResNet (default ResNet-50), ResNeXt, Wide ResNet
- EfficientNet, MobileNet, RegNet
- ConvNeXt, VGG, DenseNet
- Vision transformers: ViT, Swin, MaxViT
- All start from networks pre-trained on everyday photos

Feature classifiers (Classify)

- XGBoost (default), LightGBM, CatBoost
- Random forest and other tree models
- Logistic regression, support vector machine, small neural network

Embeddings & detection

- Embeddings: DINOv2 (default; Cell-DINO when installed), ResNet-18, ResNet-50, ConvNeXt, ViT
- YOLO object detection (behind the well detector)

Get started

```
pip install "spacr[qt]"   &&   spacr
```

Docs & tutorials	einarolafsson.github.io/spacr
Source & installers	github.com/EinarOlafsson/spacr
Preprint	bioRxiv 10.64898/2026.07.08.737057
Models	huggingface.co/einarolafsson
Example data	spacr-download --list